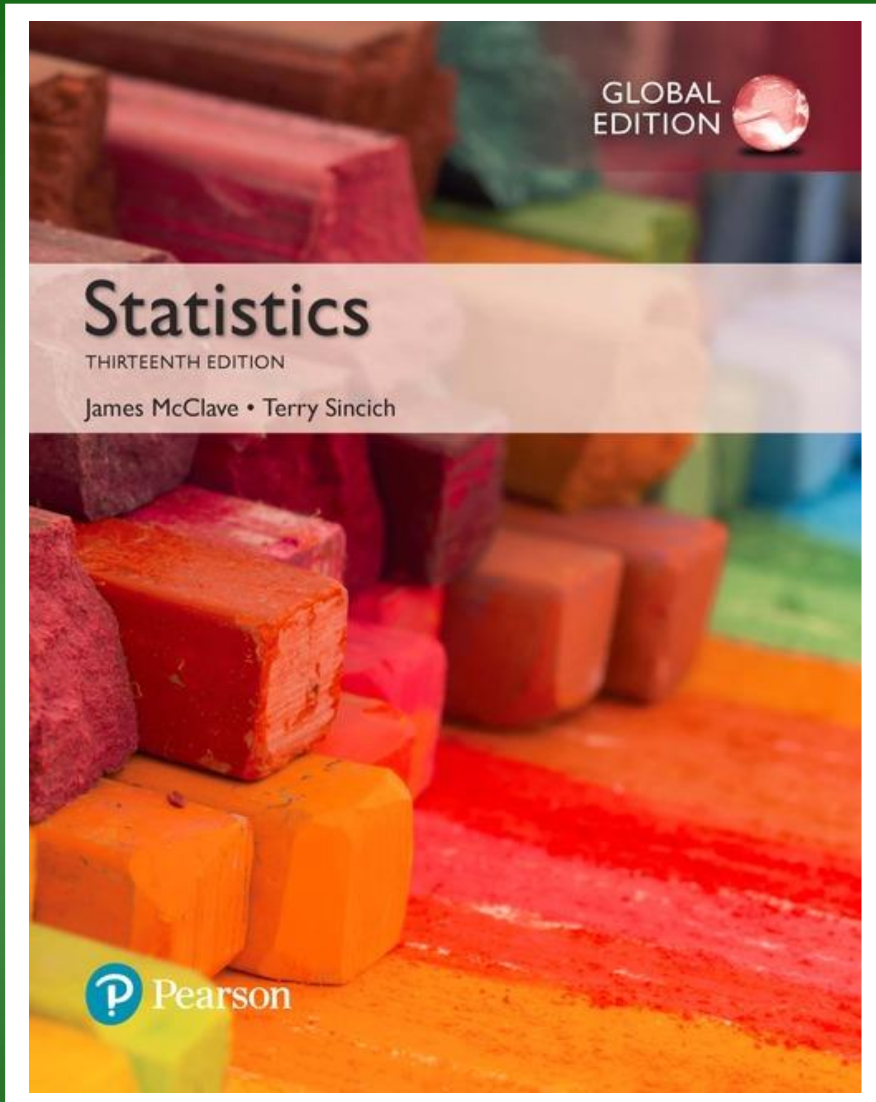


Chapter 2

Methods for Describing Sets of Data

2.1

Describing Qualitative Data

Definition

- For this study, the variable of interest, type of aphasia, is **qualitative** in nature.
- Qualitative data are nonnumerical in nature; thus, the value of a qualitative variable can only be classified into **categories** called **classes**.
- The possible types of aphasia—**Broca's**, **conduction**, and **anomic**—represent the classes for this qualitative variable.

Table 2.1 **Data on 22 Adult Aphasics**

Subject	Type of Aphasia	Subject	Type of Aphasia
1	Broca's	12	Broca's
2	Anomic	13	Anomic
3	Anomic	14	Broca's
4	Conduction	15	Anomic
5	Broca's	16	Anomic
6	Conduction	17	Anomic
7	Conduction	18	Conduction
8	Anomic	19	Broca's
9	Conduction	20	Anomic
10	Anomic	21	Conduction
11	Conduction	22	Anomic

Based on Li, E. C., Williams, S. E., and Volpe, R. D. "The effects of topic and listener familiarity of discourse variables in procedural and narrative discourse tasks." *The Journal of Communication Disorders,* Vol. 28, No. 1, Mar. 1995, p. 44 (Table 1).

A **class** is one of the categories into which qualitative data can be classified.

PEARSON

Definition

- We can summarize such data numerically in two ways:
- **(1)** by computing the **class frequency**—the number of observations in the data set that fall into each class

The **class frequency** is the number of observations in the data set that fall into a particular class.

Definition

- **(2)** by computing the class relative frequency—the proportion of the total number of observations falling into each class.

The **class relative frequency** is the class frequency divided by the total number of observations in the data set; that is,

$$\text{class relative frequency} = \frac{\text{class frequency}}{n}$$

Definition

The **class percentage** is the class relative frequency multiplied by 100; that is,

$$\text{class percentage} = (\text{class relative frequency}) \times 100$$

Definition

Summary of Graphical Descriptive Methods for Qualitative Data

Bar Graph: The categories (classes) of the qualitative variable are represented by bars, where the height of each bar is either the class frequency, class relative frequency, or class percentage.

Pie Chart: The categories (classes) of the qualitative variable are represented by slices of a pie (circle). The size of each slice is proportional to the class relative frequency.

Pareto Diagram: A bar graph with the categories (classes) of the qualitative variable (i.e., the bars) arranged by height in descending order from left to right.

2.2

Graphical Methods for Describing Quantitative Data

Definition

Summary of Graphical Descriptive Methods for Quantitative Data

Dot Plot: The numerical value of each quantitative measurement in the data set is represented by a dot on a horizontal scale. When data values repeat, the dots are placed above one another vertically.

Stem-and-Leaf Display: The numerical value of the quantitative variable is partitioned into a "stem" and a "leaf." The possible stems are listed in order in a column. The leaf for each quantitative measurement in the data set is placed in the corresponding stem row. Leaves for observations with the same stem value are listed in increasing order horizontally.

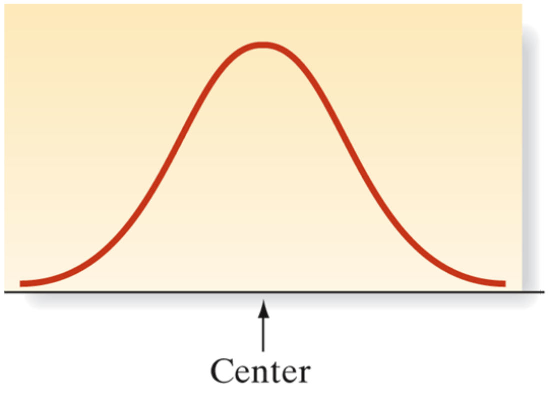
Histogram: The possible numerical values of the quantitative variable are partitioned into class intervals, each of which has the same width. These intervals form the scale of the horizontal axis. The frequency or relative frequency of observations in each class interval is determined. A vertical bar is placed over each class interval, with the height of the bar equal to either the class frequency or class relative frequency.

2.3

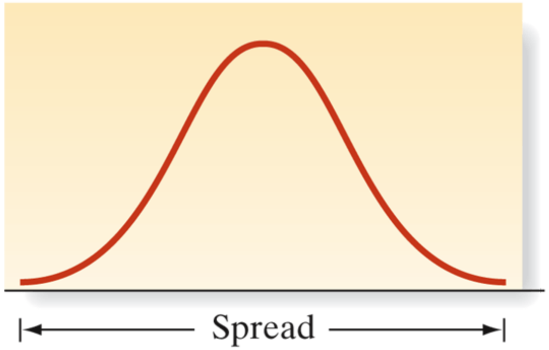
Numerical Measures of Central Tendency

Figure 2.14 Numerical descriptive measures

- Dataset → refers to either
  - sample
  - or, population
- If **statistical inference** is our goal
  - need: sample **numerical descriptive measures**
- Numerical methods to describe quantitative data measure
  - 1. The **central tendency** of the set of measurements—that is, the tendency of the data to cluster, or center, about certain numerical values.
  - 2. The **variability** of the set of measurements—that is, the spread of the data.



a.



b.

PEARSON

Definition

The most popular and best understood measure of central tendency for a quantitative data set is the ***arithmetic mean*** (or simply the **mean**) of the data set.

The **mean** of a set of quantitative data is the sum of the measurements, divided by the number of measurements contained in the data set.

Definition

- The sample mean $\bar{x}$ will play an important role in accomplishing our objective of making inferences about populations on the basis of information about the sample.
- For this reason, we need to use a different symbol for the *mean of a population*—the mean of the set of measurements on every unit in the population.
- We use the Greek letter μ (mu) for the population mean.

Symbols for the Sample Mean and the Population Mean

In this text, we adopt a general policy of using Greek letters to represent numerical descriptive measures of the population and Roman letters to represent corresponding descriptive measures of the sample. The symbols for the mean are

$$\bar{x} = \text{Sample mean} \qquad \mu = \text{Population mean}$$

PEARSON

Definition

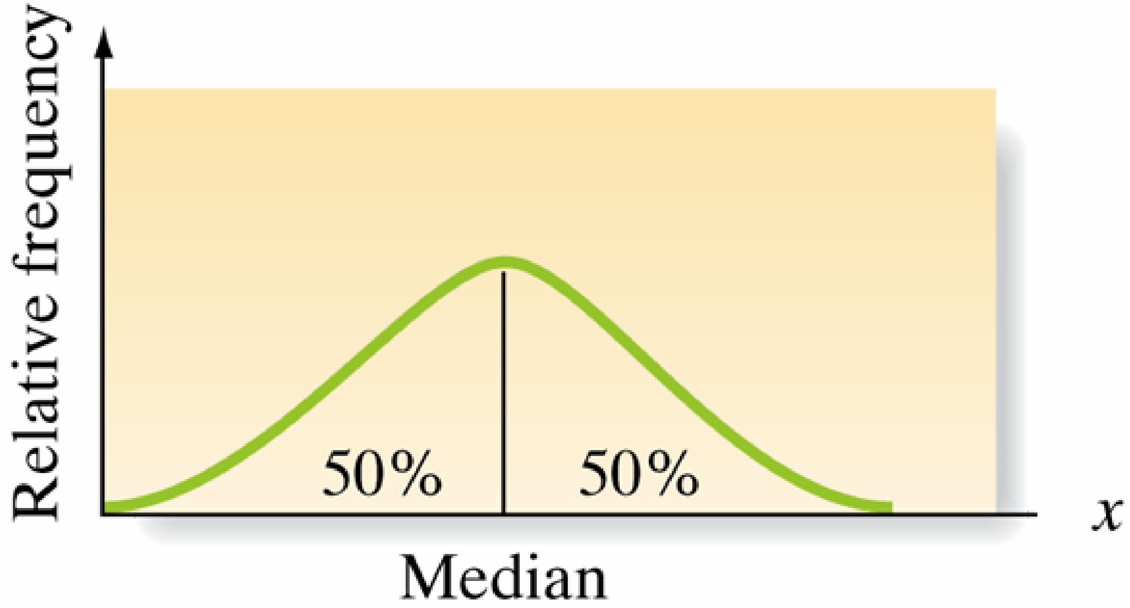
Another important measure of central tendency is the **median**.

The **median** of a quantitative data set is the middle number when the measurements are arranged in ascending (or descending) order.

PEARSON

Figure 2.16 Location of the Median

- The median is of **most value** in describing large data sets.
- If a data set is characterized by a relative frequency histogram, the median is the point on the *x*-axis such that half the area under the histogram lies above the median and half lies below.



Procedure and Symbols

We denote the *median* of a *sample* by M. Like with the population mean, we use a Greek letter (η) to represent the population median.

Calculating a Sample Median M

Arrange the n measurements from the smallest to the largest.

1. If n is odd, M is the middle number.
2. If n is even, M is the mean of the middle two numbers.

Symbols for the Sample and Population Median

$$M = \text{Sample median}$$

$$\eta = \text{Population median}$$

Definition

> A data set is said to be **skewed** if one tail of the distribution has more extreme observations than the other tail.

- A comparison of the mean and the median gives us a general method for detecting skewness in data sets.

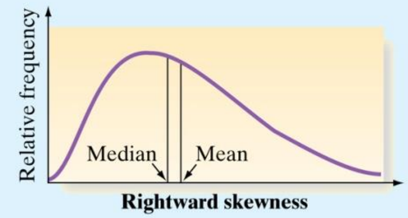
Procedure

- With ***rightward skewed*** data, the right tail (high end) of the distribution has more extreme observations.

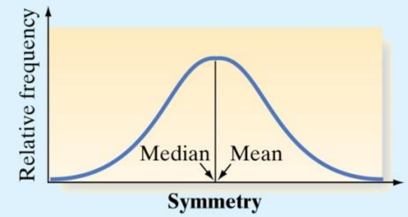
- Conversely, with ***leftward skewed*** data, the left tail (low end) of the distribution has more extreme observations.

Detecting Skewness by Comparing the Mean and the Median

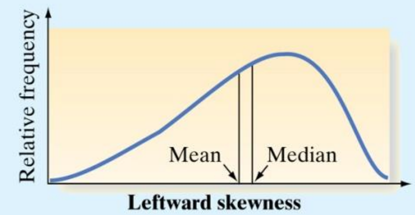
If the data set is skewed to the right, then the median is less than the mean.



If the data set is symmetric, then the mean equals the median.



If the data set is skewed to the left, then the mean is less than the median.



Definition

A third measure of central tendency is the **mode** of a set of measurements

The **mode** is the measurement that occurs most frequently in the data set.

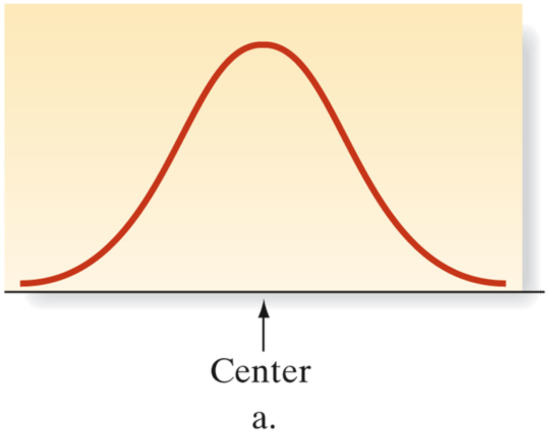
Therefore, the mode shows where the data **tend to concentrate**.

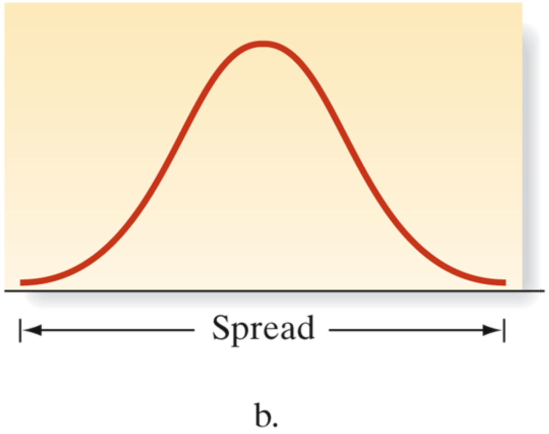
2.4

Numerical Measures of Variability

Figure 2.14 Numerical descriptive measures

- Dataset → refers to either
  - sample
  - or, population
- If **statistical inference** is our goal
  - need: sample **numerical descriptive measures**
- Numerical methods to describe quantitative data measure
  - 1. The **central tendency** of the set of measurements—that is, the tendency of the data to cluster, or center, about certain numerical values.
  - 2. The **variability** of the set of measurements—that is, the spread of the data.

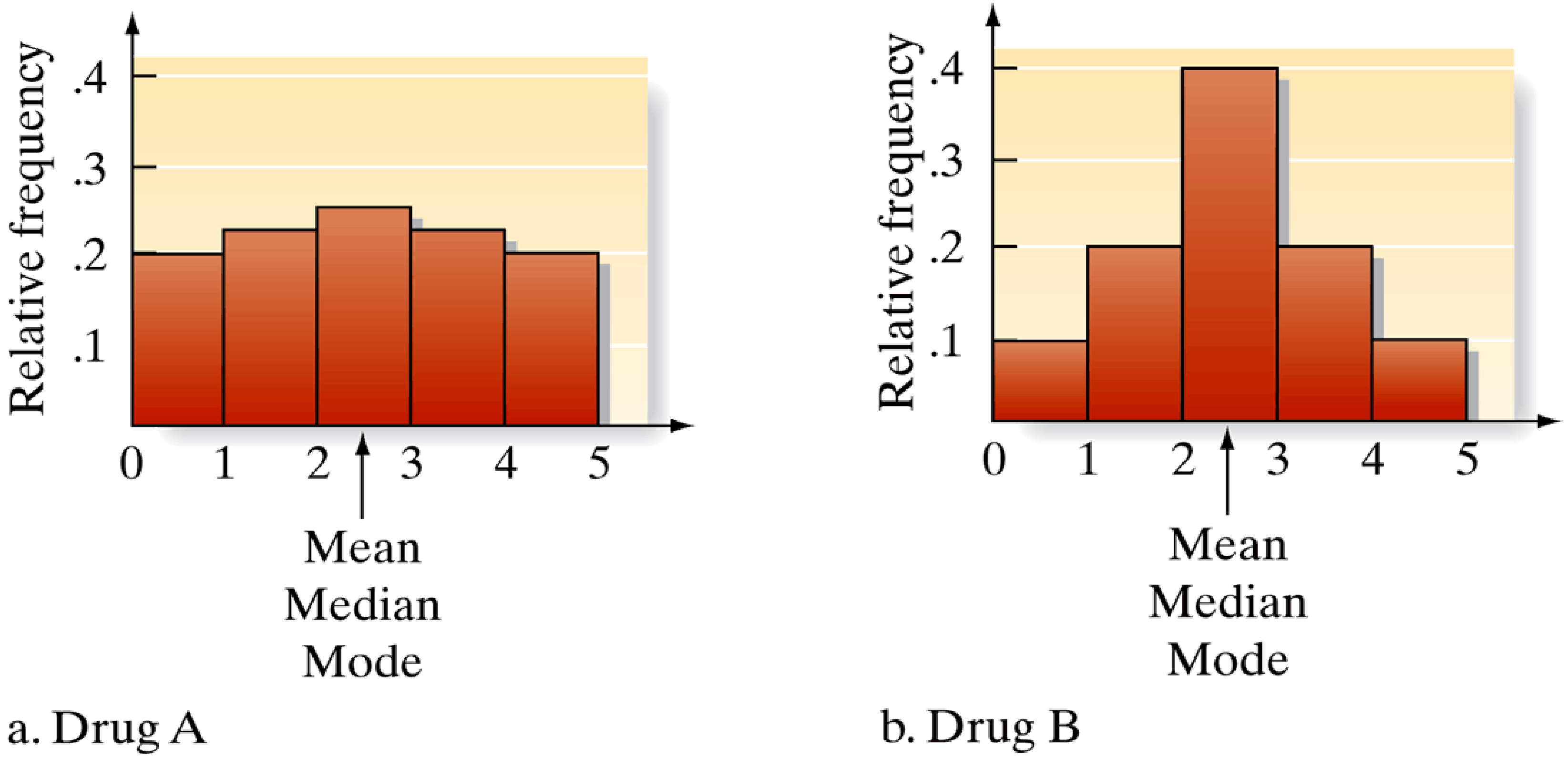




PEARSON

Figure 2.19 Response time histograms for two drugs

Same **mean**, **mode**, and **median**. Symmetric data.
For which drug, the response time is ***less variable***?



PEARSON

Definition

Perhaps the simplest measure of the variability of a quantitative data set is its ***range***.

The **range** of a quantitative data set is equal to the largest measurement minus the smallest measurement.

- The range is easy to compute and easy to understand, but it is a rather insensitive measure of data variation when the data sets are large.
- This is because two data sets can have the same range and be vastly different with respect to data variation.

Let's see if we can find a measure of data variation that is more sensitive than the range.

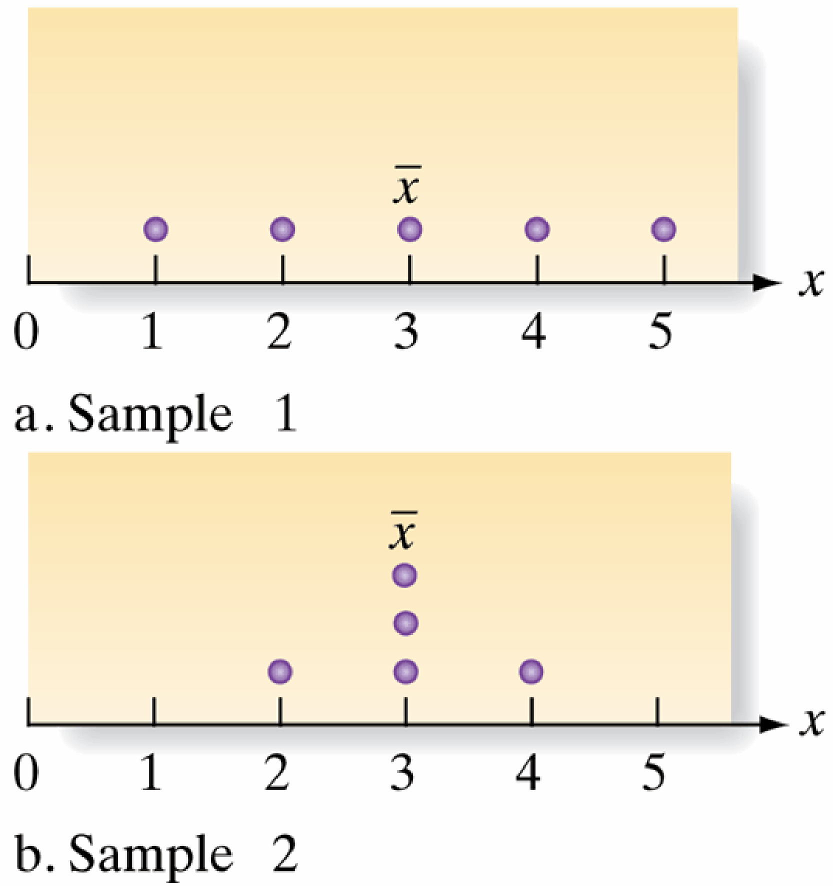
Table 2.5 Two Hypothetical Data Sets

	Sample 1	Sample 2
Measurements	1, 2, 3, 4, 5	2, 3, 3, 3, 4
Mean	$\bar{x} = \frac{1 + 2 + 3 + 4 + 5}{5} = \frac{15}{5} = 3$	$\bar{x} = \frac{2 + 3 + 3 + 3 + 4}{5} = \frac{15}{5} = 3$
Deviations of measurement values from $\bar{x}$	$(1 - 3), (2 - 3), (3 - 3), (4 - 3), (5 - 3)$ or $-2, -1, 0, 1, 2$	$(2 - 3), (3 - 3), (3 - 3), (3 - 3), (4 - 3)$ or $-1, 0, 0, 0, 1$

- If they tend to be large in magnitude, as in sample 1, the data are spread out, or highly variable.
- If the deviations are mostly small, as in sample 2, the data are clustered around the mean, and therefore do not exhibit much variability

Figure 2.20 Dot plots for deviations in Table 2.5

- You can see that these deviations provide information about the **variability** of the sample measurements.
- The next step is to condense the information in these distances into a single numerical measure of variability.
- Averaging the deviations from $\bar{x}$ won't help because the negative and positive deviations cancel.



Definition

To use the squared deviations calculated from a data set, we first calculate the ***sample variance***.

The **sample variance** for a sample of n measurements is equal to the sum of the squared deviations from the mean, divided by $(n - 1)$. The symbol s^2 is used to represent the sample variance.

Formula

Formula for the Sample Variance:

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1}$$

Note: A shortcut formula for calculating s^2 is

$$s^2 = \frac{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}{n - 1}$$

Example

Table 2.5 Two Hypothetical Data Sets

	Sample 1	Sample 2
Measurements	$1, 2, 3, 4, 5$	$2, 3, 3, 3, 4$
Mean	$\bar{x} = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$	$\bar{x} = \frac{2+3+3+3+4}{5} = \frac{15}{5} = 3$
Deviations of measurement values from $\bar{x}$	$(1-3), (2-3), (3-3), (4-3), (5-3)$ or $-2, -1, 0, 1, 2$	$(2-3), (3-3), (3-3), (3-3), (4-3)$ or $-1, 0, 0, 0, 1$

Calculate the variance for sample 1.

$$s^2 = \frac{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2}{5-1}$$

$$= \frac{4+1+0+1+4}{4} = 2.5$$

PEARSON

Definition

The second step in finding a meaningful measure of data variability is to calculate the ***standard deviation*** of the data set.

The **sample standard deviation**, s, is defined as the positive square root of the sample variance, s^2, or, mathematically,

$$s = \sqrt{s^2}$$

Definition

The population variance, denoted by the symbol σ^2(sigma squared), is the average of the squared deviations from the mean, μ, of the measurements on all units in the population, and σ (sigma) is the square root of this quantity.

Symbols for Variance and Standard Deviation

s^2 = Sample variance

s = Sample standard deviation

σ^2 = Population variance

σ = Population standard deviation

Notice that, unlike the variance, the standard deviation is expressed in the original units of measurement.

Example

Problem Calculate the variance and standard deviation of the following sample: 2, 3, 3, 3, 4.

Solution

Table 2.6 Calculating s^2		
x	$(x - \bar{x})$	$(x - \bar{x})^2$
2	−1	1
3	0	0
3	0	0
3	0	0
4	1	1
$\Sigma x = 15$		$\Sigma(x - \bar{x})^2 = 2$

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} = \frac{2}{5 - 1} = \frac{2}{4} = .5$$

$$s = \sqrt{.5} = .71$$

Figure 2.21 SAS numerical descriptive measures for 100 EPA mileages

The MEANS Procedure

Analysis Variable : MPG

Mean	Std Dev	Variance	N	Minimum	Maximum	Median
36.9940000	2.4178971	5.8462263	100	30.0000000	44.9000000	37.0000000

- You now know that the standard deviation measures the variability of a set of data, and you know how to calculate the standard deviation.
- The larger the standard deviation, the more variable the data are.
- The smaller the standard deviation, the less variation there is in the data.
- But how can we practically interpret the standard deviation and use it to make inferences? This is the topic of next week.

2.5

Using the Mean and Standard Deviation to Describe Data

Rule 2.1

Rule 2.1 Using the Mean and Standard Deviation to Describe Data: Chebyshev's Rule

Chebyshev's rule applies to any data set, regardless of the shape of the frequency distribution of the data.

a. It is possible that very few of the measurements will fall within one standard deviation of the mean [i.e., within the interval $(\bar{x} - s, \bar{x} + s)$ for samples and $(\mu - \sigma, \mu + \sigma)$ for populations].

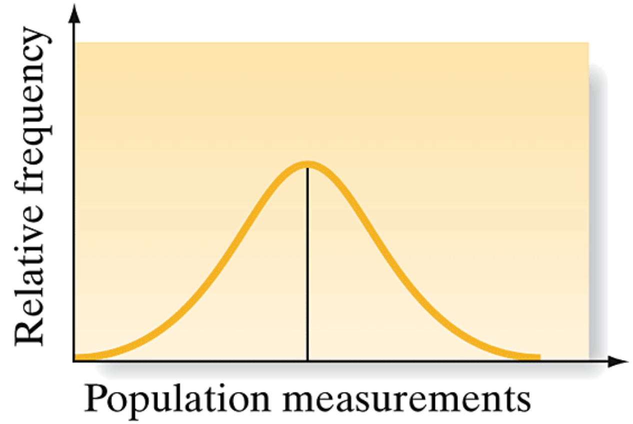
b. At least $\frac{3}{4}$ of the measurements will fall within two standard deviations of the mean [i.e., within the interval $(\bar{x} - 2s, \bar{x} + 2s)$ for samples and $(\mu - 2\sigma, \mu + 2\sigma)$ for populations].

c. At least $\frac{8}{9}$ of the measurements will fall within three standard deviations of the mean [i.e., within the interval $(\bar{x} - 3s, \bar{x} + 3s)$ for samples and $(\mu - 3\sigma, \mu + 3\sigma)$ for populations].

d. Generally, for any number k greater than 1, at least $(1 - 1/k^2)$ of the measurements will fall within k standard deviations of the mean [i.e., within the interval $(\bar{x} - ks, \bar{x} + ks)$ for samples and $(\mu - k\sigma, \mu + k\sigma)$ for populations].

Rule 2.2 Using the Mean and Standard Deviation to Describe Data: The Empirical Rule

The **empirical rule** is a rule of thumb that applies to data sets with frequency distributions that are mound shaped and symmetric, as follows:



a. Approximately 68% of the measurements will fall within one standard deviation of the mean [i.e., within the interval $(\bar{x} - s, \bar{x} + s)$ for samples and $(\mu - \sigma, \mu + \sigma)$ for populations].

b. Approximately 95% of the measurements will fall within two standard deviations of the mean [i.e., within the interval $(\bar{x} - 2s, \bar{x} + 2s)$ for samples and $(\mu - 2\sigma, \mu + 2\sigma)$ for populations].

c. Approximately 99.7% (essentially all) of the measurements will fall within three standard deviations of the mean [i.e., within the interval $(\bar{x} - 3s, \bar{x} + 3s)$ for samples and $(\mu - 3\sigma, \mu + 3\sigma)$ for populations].

Example

Problem Thirty students in an experimental psychology class use various techniques to train a rat to move through a maze. At the end of the course, each student's rat is timed as it negotiates the maze. The results (in minutes) are listed in Table 2.7. Determine the fraction of the 30 measurements in the intervals $\bar{x} \mp s, \bar{x} \mp s2$, *and* $\bar{x} \mp 3s$, and compare the results with those predicted using Rules 2.1 and 2.2.

Table 2.7 Times (in Minutes) of 30 Rats Running through a Maze

1.97	.60	4.02	3.20	1.15	6.06	4.44	2.02	3.37	3.65
1.74	2.75	3.81	9.70	8.29	5.63	5.21	4.55	7.60	3.16
3.77	5.36	1.06	1.71	2.47	4.25	1.93	5.15	2.06	1.65

PEARSON

Solution

First, we entered the data into the computer and used MINITAB to produce summary statistics. The mean and standard deviation of the sample data, highlighted on the printout shown in Figure 2.22, are

$$\bar{x} = 3.74 \text{ minutes} \quad s = 2.20 \text{ minutes}$$

Descriptive Statistics: RUNTIME

Variable	N	Mean	StDev	Minimum	Median	Maximum
RUNTIME	30	3.744	2.198	0.600	3.510	9.700

1st interval: $(\bar{x} - s, \bar{x} + s) = (3.74 - 2.20, 3.74 + 2.20) = (1.54, 5.94)$

A check of the measurements shows that 23 of the times are within this one standard deviation interval around the mean. This number represents **23/30, or ≈ 77%,** of the sample measurements.

Solution

These one-, two-, and three-standard-deviation percentages (77%, 93%, and 100%) agree fairly well with the approximations of 68%, 95%, and 100% given by the empirical rule (Rule 2.2).

First, we entered the data into the computer and used MINITAB to produce summary statistics. The mean and standard deviation of the sample data, highlighted on the printout shown in Figure 2.22, are

$$\bar{x} = 3.74 \text{ minutes} \quad s = 2.20 \text{ minutes}$$

Descriptive Statistics: RUNTIME

Variable	N	Mean	StDev	Minimum	Median	Maximum
RUNTIME	30	3.744	2.198	0.600	3.510	9.700

2nd interval: $(\bar{x} - 2s, \bar{x} + 2s) = (3.74 - 4.40, 3.74 + 4.40) = (-.66, 8.14)$

All but two of the times are within this interval, so 28/30, or approximately 93%, are within two standard deviations of $\bar{x}$.

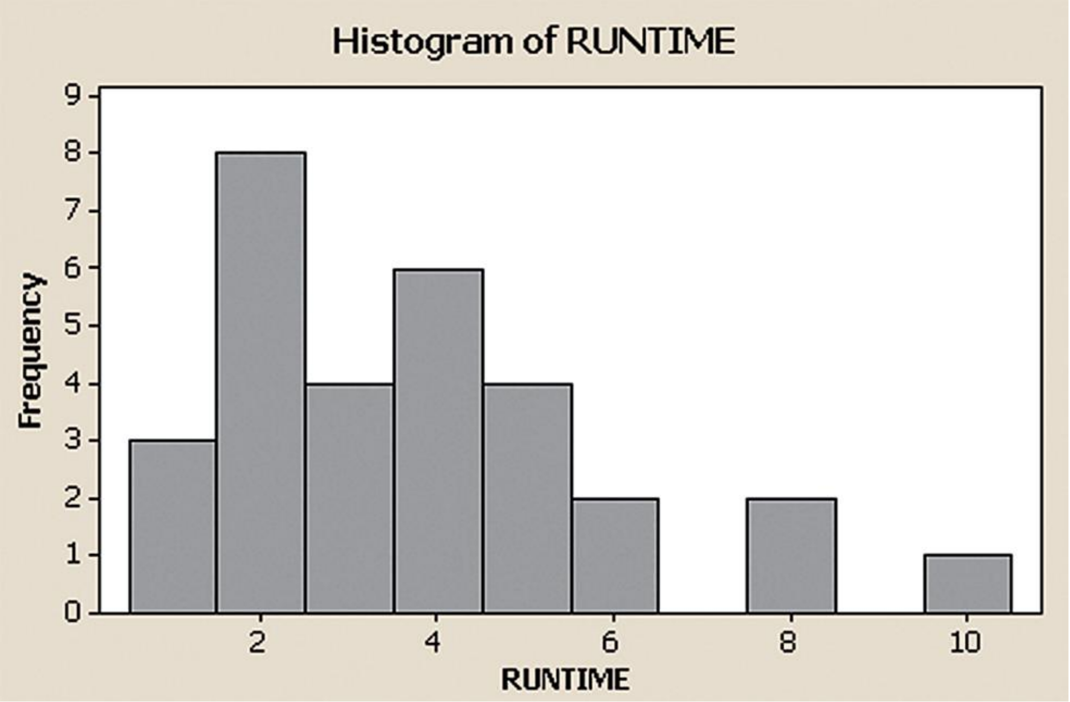
3rd interval: $(\bar{x} - 3s, \bar{x} + 3s) = (3.74 - 6.60, 3.74 + 6.60) = (-2.86, 10.34)$

All of the times fall within three standard deviations of the mean.

Figure 2.23 MINITAB histogram of rat maze times

If you look at the MINITAB frequency histogram for this data set (Figure 2.23), you'll note that the distribution is not really mound shaped, nor is it extremely skewed. Thus, we get reasonably good results from the mound-shaped approximations.

Of course, we know from Chebyshev's rule (Rule 2.1) that no matter what the shape of the distribution, we would expect at least 75% and at least 89% of the measurements to lie within two and three standard deviations of $\bar{x}$, respectively.



PEARSON

Example

Problem Chebyshev's rule and the empirical rule are useful as a check on the calculation of the standard deviation. For example, suppose we calculated the standard deviation for the gas mileage data to be **5.85**. Are there any "clues" in the data that enable us to judge whether this number is reasonable?

Table 2.2 EPA Mileage Ratings on 100 Cars

36.3	41.0	36.9	37.1	44.9
32.7	37.3	41.2	36.6	32.9
40.5	36.5	37.6	33.9	40.2
36.2	37.9	36.0	37.9	35.9
38.5	39.0	35.5	34.8	38.6
36.3	36.8	32.5	36.4	40.5
41.0	31.8	37.3	33.1	37.0
37.0	37.2	40.7	37.4	37.1
37.1	40.3	36.7	37.0	33.9
39.9	36.9	32.9	33.8	39.8
36.8	30.0	37.2	42.1	36.7
36.5	33.2	37.4	37.5	33.6
36.4	37.7	37.7	40.0	34.2
38.2	38.3	35.7	35.6	35.1
39.4	35.3	34.4	38.8	39.7
36.6	36.1	38.2	38.4	39.3
37.6	37.0	38.7	39.0	35.8
37.8	35.9	35.6	36.7	34.5
40.1	38.0	35.2	34.8	39.5
34.0	36.8	35.0	38.1	36.9

PEARSON

Solution

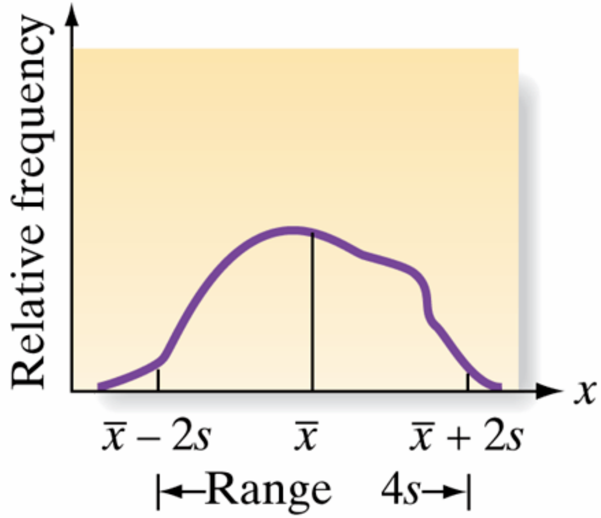
- The range of the mileage data is $44.9 - 30.0 = 14.9$.
- From Chebyshev's rule and the empirical rule, we know that most of the measurements (approximately 95% if the distribution is mound shaped) will be within two standard deviations of the mean.
- And regardless of the shape of the distribution and the number of measurements, almost all of them will fall within three standard deviations of the mean.
- Consequently, we would expect the range of the measurements to be between 4 and 6 standard deviations in length.
- For the car mileage data, this means that s should fall between

$$\frac{\text{Range}}{6} = \frac{14.9}{6} = 2.48 \text{ and } \frac{\text{Range}}{4} = \frac{14.9}{4} = 3.73$$

PEARSON

Figure 2.24 The relation between the range and the standard deviation

- Hence, the standard deviation should not be much larger than 1/4 of the range, particularly for the data set with 100 measurements. Thus, we have reason to believe that the calculation of **5.85 is too large**. A check of our work reveals that 5.85 is the variance s^2, not the standard deviation s.
- We "forgot" to take the square root (a common error); the correct value is $s = 2.42$.
- Note that this value is slightly smaller than the range divided by 6 (2.48).
- The larger the data set, the greater is the tendency for very large or very small measurements (extreme values) to appear, and when they do, the range may exceed six standard deviations.



PEARSON

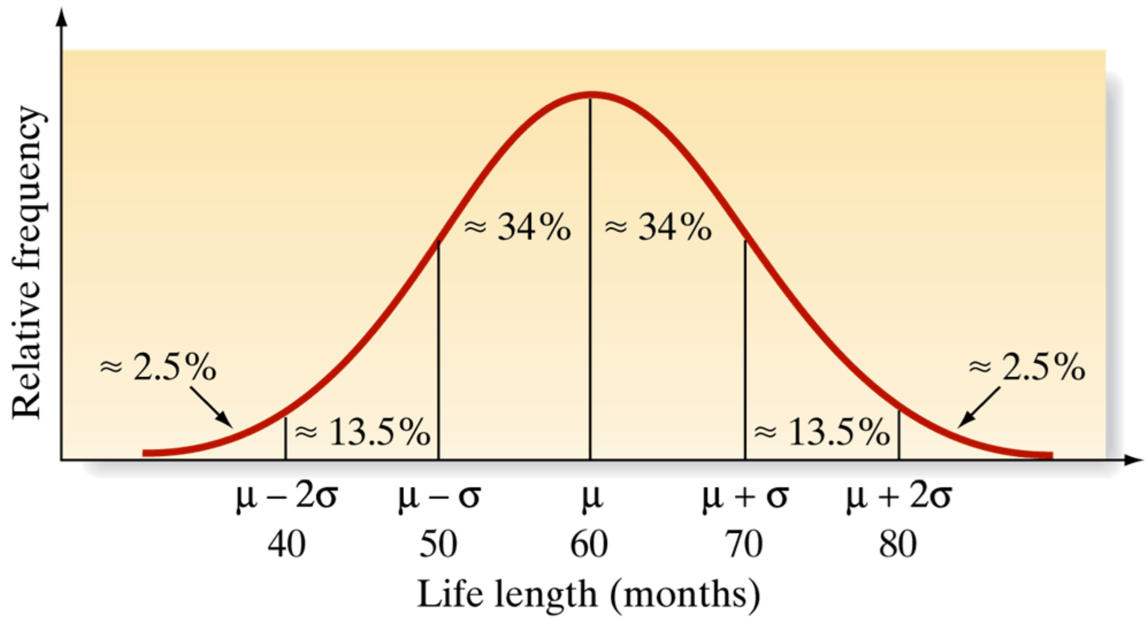
Example

Problem A manufacturer of automobile batteries claims that the average length of life for its grade A battery is 60 months. However, the guarantee on this brand is for just 36 months. Suppose the standard deviation of the life length is known to be 10 months and the frequency distribution of the life-length data is known to be mound shaped.

a) Approximately what percentage of the manufacturer's grade A batteries will last more than 50 months, assuming that the manufacturer's claim is true?
b) Approximately what percentage of the manufacturer's batteries will last less than 40 months, assuming that the manufacturer's claim is true?
c) Suppose your battery lasts 37 months. What could you infer about the manufacturer's claim?

Solution - Figure 2.25 Battery life-length distribution: manufacturer's claim assumed true

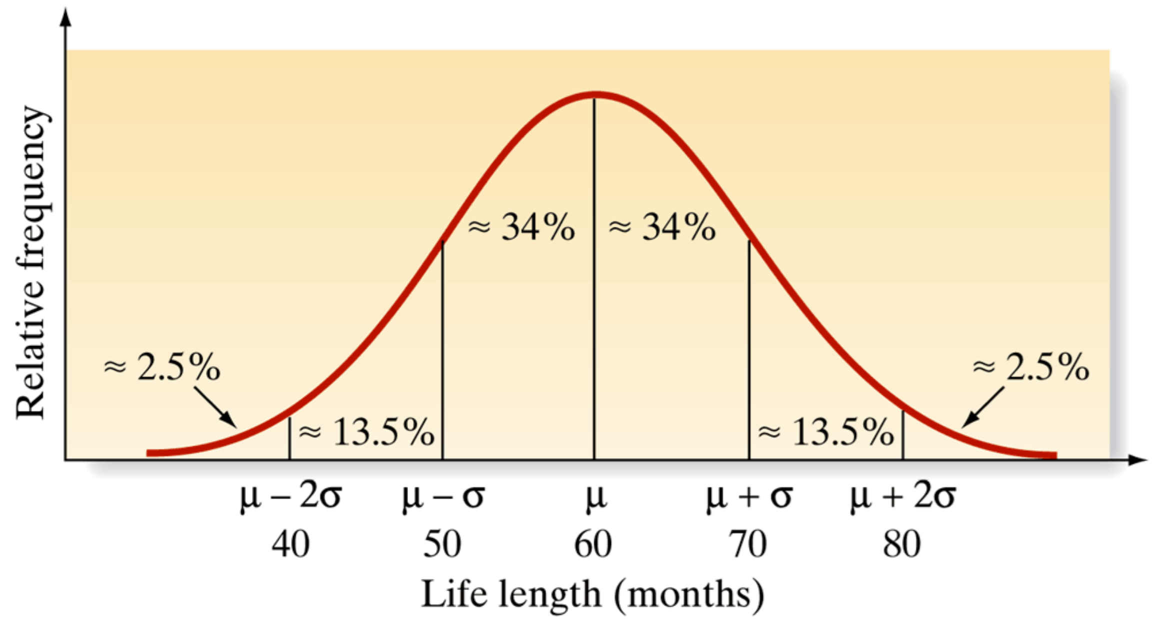
If the distribution of life length is assumed to be mound shaped with a mean of 60 months and a standard deviation of 10 months, it would appear as shown in Figure 2.25. Note that we can take advantage of the fact that mound-shaped distributions are (approximately) symmetric about the mean, so that the percentages given by the empirical rule can be split equally between the halves of the distribution on each side of the mean.



PEARSON

Solution

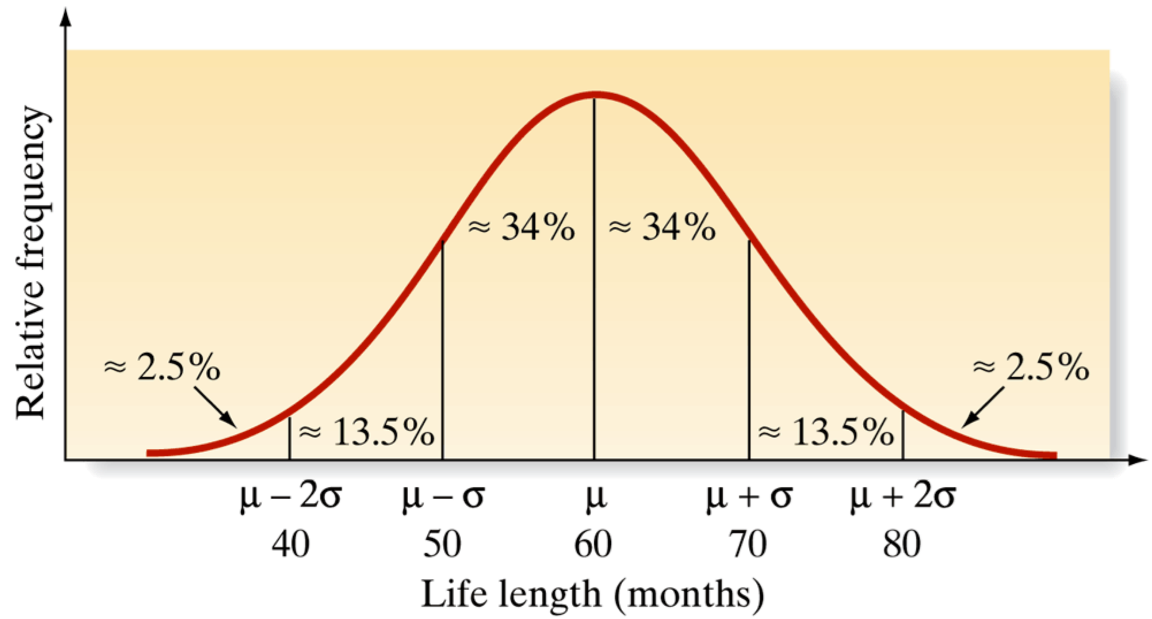
a) It is easy to see in Figure 2.25 that the percentage of batteries lasting more than 50 months is approximately 34% (between 50 and 60 months) plus 50% (greater than 60 months). Thus, approximately 84% of the batteries should have a life exceeding 50 months.



PEARSON

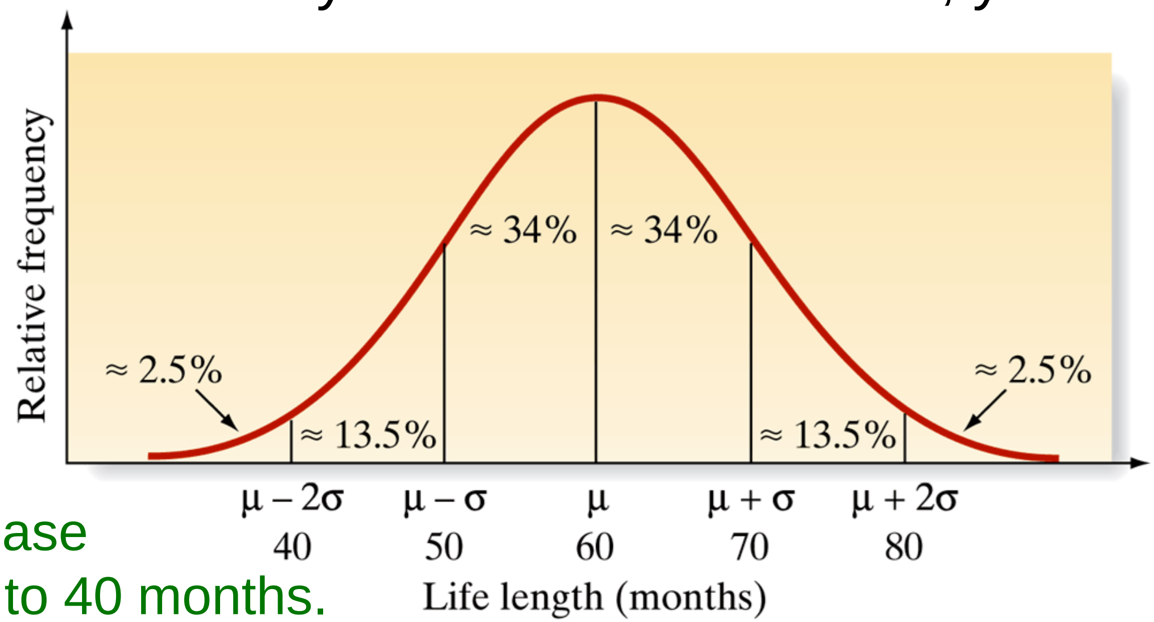
Solution

b) The percentage of batteries that last less than 40 months can also be easily determined from Figure 2.25: Approximately 2.5% of the batteries should fail prior to 40 months, assuming that the manufacturer's claim is true.



Solution

c) If you are so unfortunate that your grade A battery fails at 37 months, you can make one of two inferences: Either your battery was one of the approximately **2.5% that fail** prior to 40 months, or something about the manufacturer's **claim is not true**. Because the chances are so small that a battery fails before 40 months, you would have good reason to have serious doubts about the manufacturer's claim.



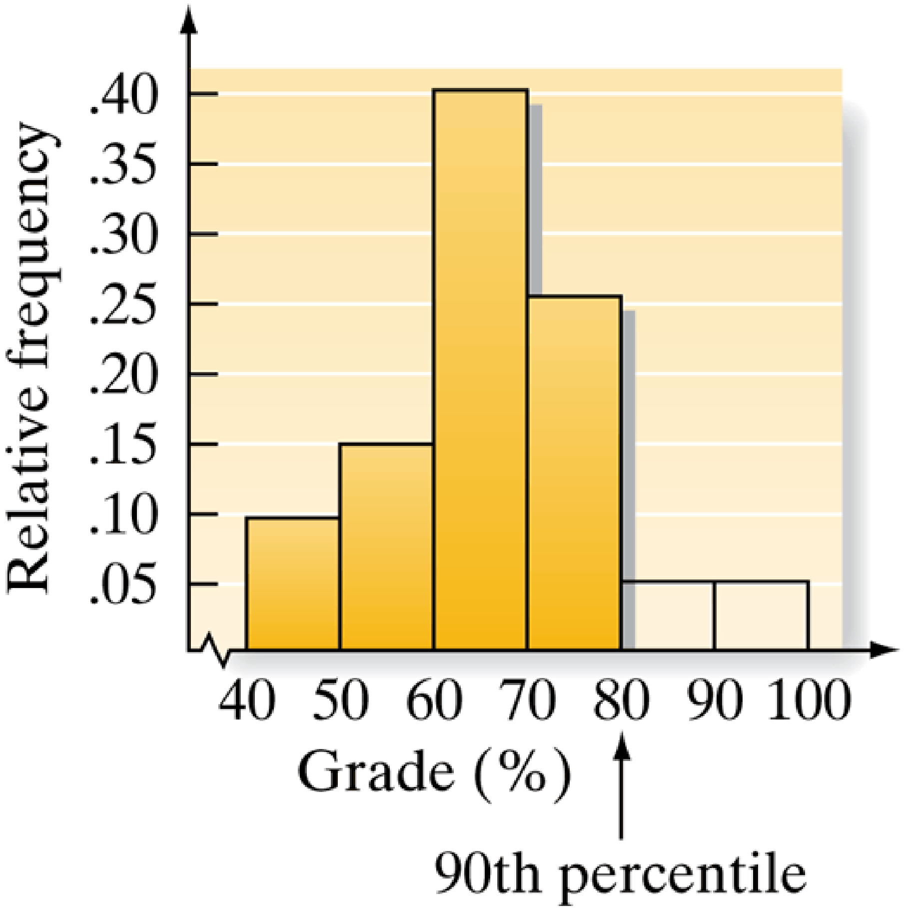
A mean smaller than 60 months or a standard deviation longer than 10 months would each increase the likelihood of failure prior to 40 months.

2.6

Numerical Measures of Relative Standing

Figure 2.26 Location of 90th percentile for test grades

One measure of the relative standing of a measurement is its **percentile ranking**, or **percentile score**.
For example, suppose you scored an 80 on a test and you want to know how you fared in comparison with others. If you scored at the 90th percentile, it means that 90% of the grades were lower than yours and 10% were higher.

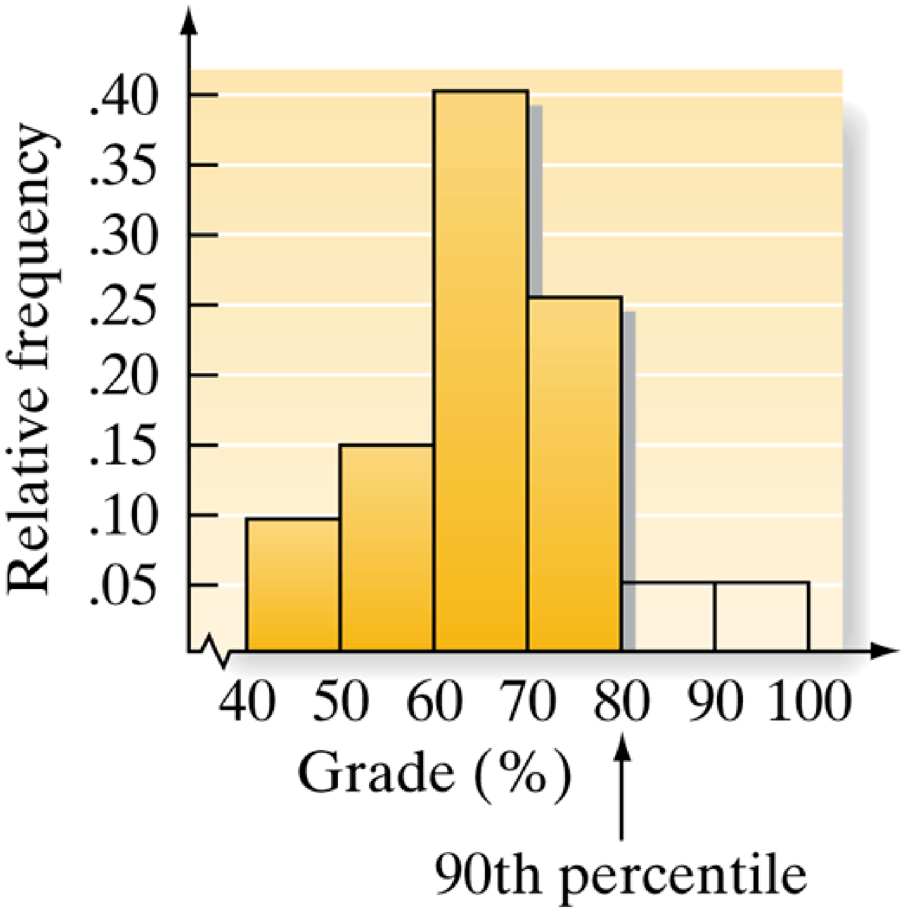


PEARSON

Figure 2.26 Location of 90th percentile for test grades

Thus, if the scores were described by the relative frequency histogram in Figure 2.26, the 90th percentile would be located at a point such that 90% of the total area under the relative frequency histogram lies below the 90th percentile and 10% lies above.

If you scored in the 50th percentile, 50% of the test grades would be lower than yours and 50% would be higher.



PEARSON

Definition

- Percentile rankings are of practical value only with large data sets.
- Finding them involves a process similar to the one used in finding a median.
- The measurements are ranked in order, and a rule is selected to define the location of each percentile.
- Since we are interested primarily in interpreting the percentile rankings of measurements (rather than in finding particular percentiles for a data set), we define the *p*th *percentile* of a data set as follows.

For any set of n measurements (arranged in ascending or descending order), the ***p*th percentile** is a number such that $p\%$ of the measurements fall below that number and $(100 - p)\%$ fall above it.

PEARSON

Example - Figure 2.27 SPSS percentile statistics for water-level task deviations

Problem Refer to the water-level task deviations for the 40 subjects in Table 2.4. An SPSS printout describing the data is shown in Figure 2.27. Locate the 25th percentile and 95th percentile on the printout and interpret the associated values.

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Deviation	40	100.0%	0	.0%	40	100.0%

Percentiles

		Percentiles						
		5	10	25	50	75	90	95
Weighted Average (Definition 1)	Deviation	-4.90	-1.00	2.00	7.00	10.00	15.80	24.80
Tukey's Hinges	Deviation			2.00	7.00	10.00		

PEARSON

Solution

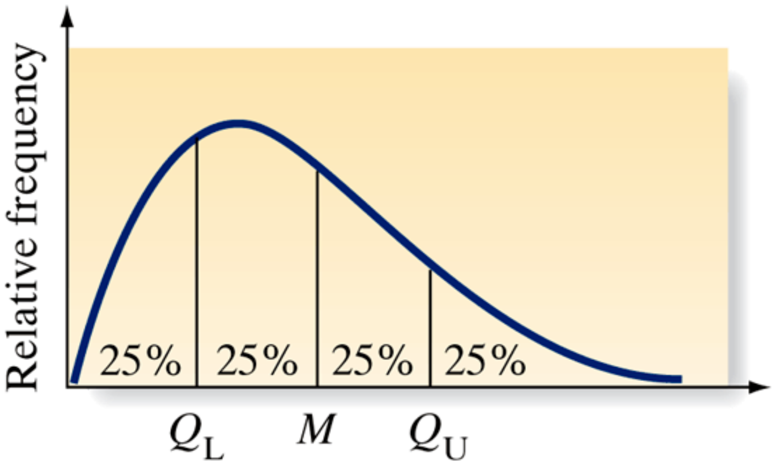
Both the 25th percentile and 95th percentile are highlighted on the SPSS printout. The values in question are 2.0 and 24.8, respectively. Our interpretations are as follows: **25% of the 40 deviations fall below 2.0** and **95% of the deviations fall below 24.8**.

Percentiles

		Percentiles						
		5	10	25	50	75	90	95
Weighted Average (Definition 1)	Deviation	-4.90	-1.00	2.00	7.00	10.00	15.80	24.80
Tukey's Hinges	Deviation			2.00	7.00	10.00		

Figure 2.28 The quartile for a data set

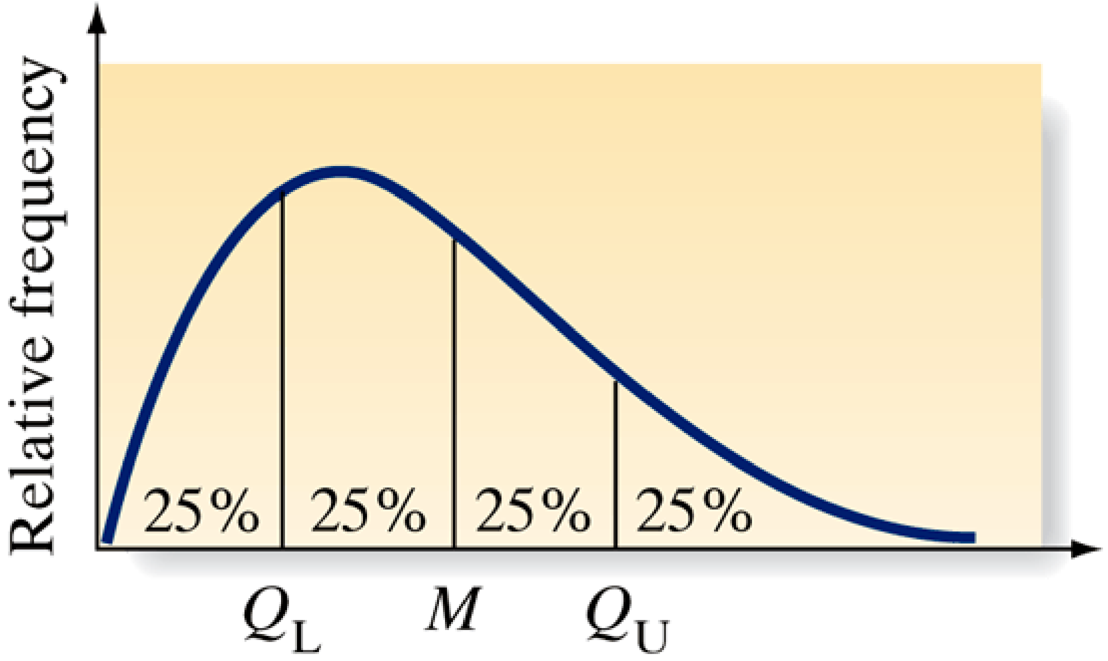
- Percentiles that partition a data set into four categories, each category containing exactly 25% of the measurements, are called **quartiles**.
- The *lower quartile* (Q_L) is the 25th percentile, the *middle quartile* (*M*) is the median or 50th percentile, and the *upper quartile* (Q_U) is the 75th percentile, as shown in Figure 2.28.
- Quartiles will prove useful in finding unusual observations in a data set



PEARSON

Definition

The **lower quartile (Q_L)** is the 25th percentile of a data set. The **middle quartile (M)** is the median or 50th percentile. The **upper quartile (Q_U)** is the 75th percentile.



Formula

Another measure of relative standing in popular use is the ***z*-score**. As you can see in the following definition, the *z*-score makes use of the mean and standard deviation of a data set in order to specify the relative location of the measurement.

The **sample** z**-score** for a measurement x is

$$z = \frac{x - \bar{x}}{s}$$

The **population** z**-score** for a measurement x is

$$z = \frac{x - \mu}{\sigma}$$

Example

Problem Suppose a sample of 2,000 high school seniors' verbal SAT scores is selected. The mean and standard deviation are

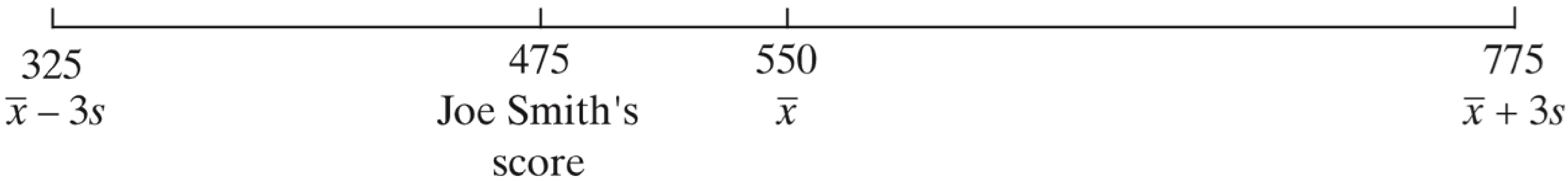
$$\bar{x} = 550 \qquad s = 75$$

Suppose Joe Smith's score is 475.

What is his sample *z*-score?

Solution - Figure 2.29 Verbal SAT scores of high school seniors

Solution Joe Smith’s verbal SAT score lies below the mean score of the 2,000 seniors, as shown in Figure 2.29.



We compute

$$z = \frac{x - \bar{x}}{s} = \frac{475 - 550}{75} = -1.0$$

which tells us that Joe Smith’s score is 1.0 standard deviation below the sample mean; in short, his sample z-score is -1.0.

PEARSON

Definition

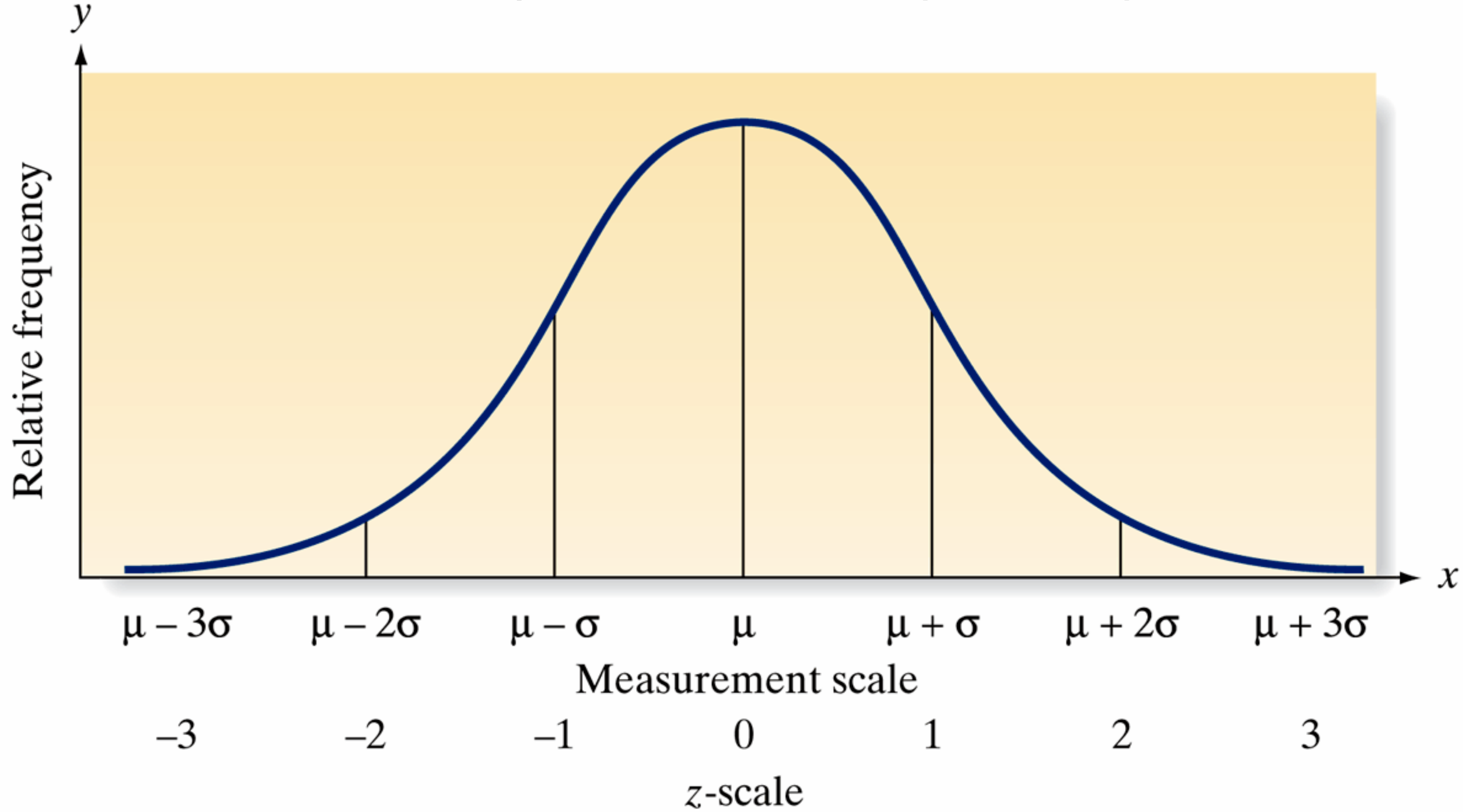
We can be more specific if we know that the frequency distribution of the measurements is mound shaped. In this case, the following interpretation of the *z*-score can be given:

Interpretation of *z*-Scores for Mound-Shaped Distributions of Data

1. Approximately 68% of the measurements will have a z-score between -1 and 1.
2. Approximately 95% of the measurements will have a z-score between -2 and 2.
3. Approximately 99.7% (almost all) of the measurements will have a z-score between -3 and 3.

Figure 2.30 Population *z*-scores for a mound–shaped distribution

Note that this interpretation of *z*-scores is identical to that given by the empirical rule for mound-shaped distributions (Rule 2.2).



PEARSON

2.7

Methods for Detecting Outliers: Box plots and *z*-Scores

Definition

Sometimes it is important to identify **inconsistent** or **unusual** measurements in a data set.

An observation (or measurement) that is unusually large or small relative to the other values in a data set is called an **outlier**. Outliers typically are attributable to one of the following causes:

1. The measurement is observed, recorded, or entered into the computer incorrectly.
2. The measurement comes from a different population.
3. The measurement is correct, but represents a rare (chance) event.

Definition

Two useful methods for detecting outliers, one graphical and one numerical, are **box plots** and ***z*-scores**. The box plot is based on the quartiles of a data set. Specifically, a box plot is based on the **interquartile range** (IQR).

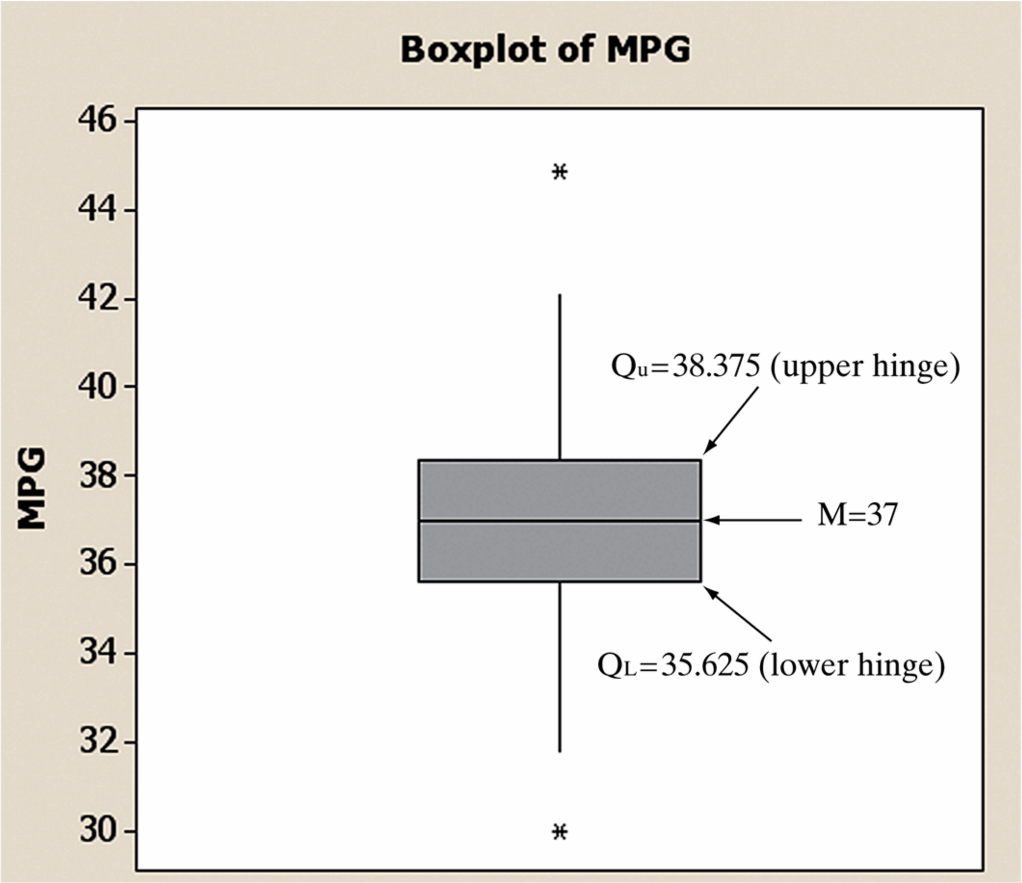
The **interquartile range (IQR)** is the distance between the lower and upper quartiles:

$$\text{IQR} = Q_U - Q_L$$

Figure 2.31 Annotated MINITAB box plot for EPA gas mileages

A rectangle (the *box*) is drawn, with the bottom and top of the rectangle (the **hinges**) drawn at the quartiles Q_L and Q_U, respectively.

$$IQR = 38.375 - 35.625 = 2.75$$



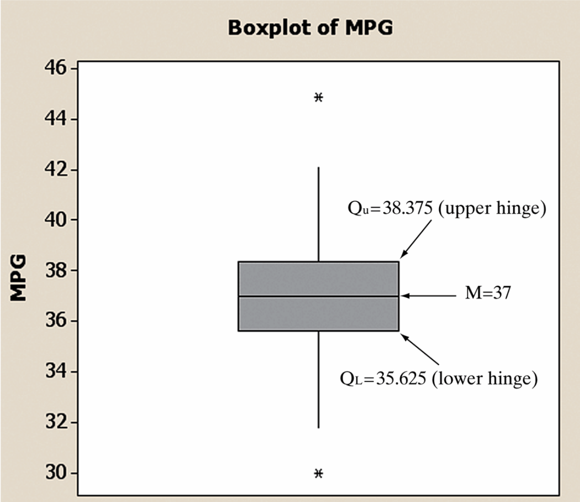
PEARSON

Box plot elements

To guide the construction of the “tails” of the box plot, two sets of limits, called **inner fences** and **outer fences**, are used. Neither set of fences actually appears on the plot. Inner fences are located at a distance of **1.5(IQR)** from the hinges. Emanating from the hinges of the box are vertical lines called the whiskers. The two whiskers extend to the most extreme observation inside the inner fences.

$$
\begin{aligned}
Lower\ inner\ fence &= Lower\ hinge - 1.5(IQR) \\
&= 35.625 - 1.5(2.75) \\
&= 35.625 - 4.125 = 31.5
\end{aligned}
$$

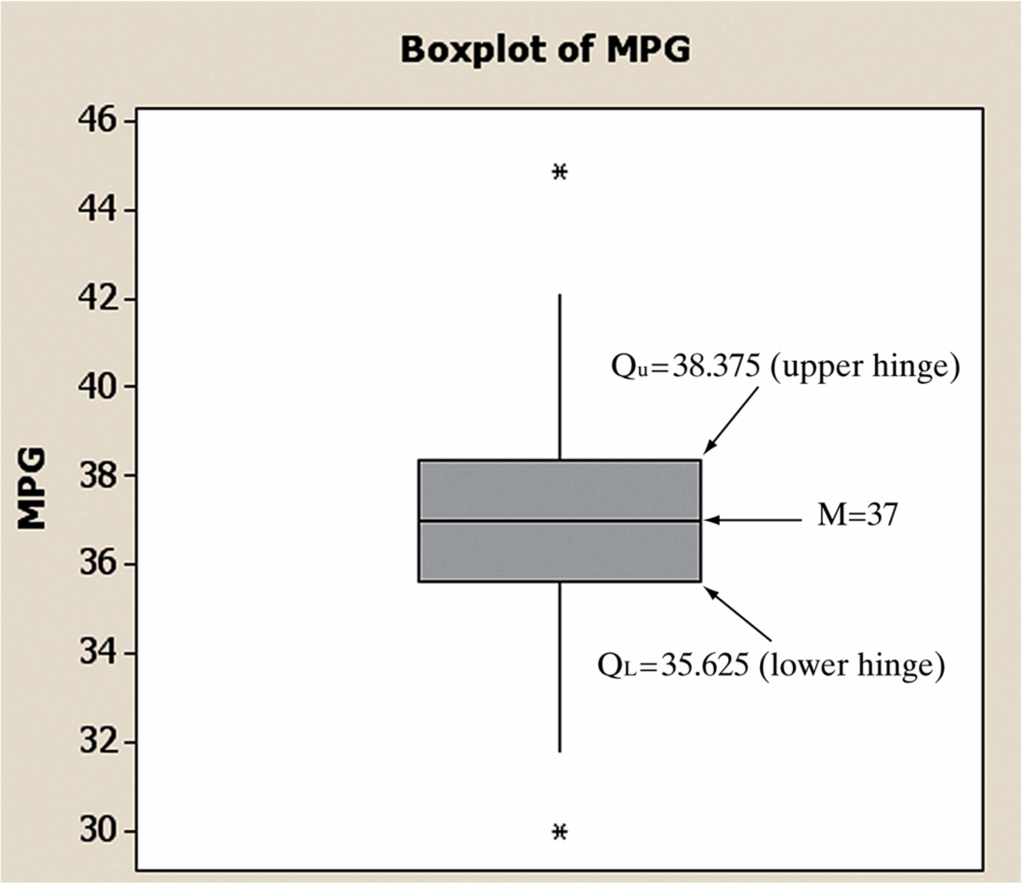
$$
\begin{aligned}
Upper\ inner\ fence &= Upper\ hinge + 1.5(IQR) \\
&= 38.375 + 1.5(2.75) \\
&= 38.375 + 4.125 = 42.5
\end{aligned}
$$



PEARSON

Box plot elements

- Values that are beyond the inner fences are deemed ***potential outliers***.
- Values that are beyond the outer fences ($3 \times IQR$) are deemed ***outliers***.
- There are **no outliers** but **two potential outliers** in this boxplot.



PEARSON

Definition

Elements of a Box Plot†

1. A rectangle (the **box**) is drawn with the ends (the **hinges**) drawn at the lower and upper quartiles (Q_L and Q_U). The median M of the data is shown in the box, usually by a line or a symbol (such as "+").
2. The points at distances 1.5(IQR) from each hinge mark the **inner fences** of the data set. Lines (the **whiskers**) are drawn from each hinge to the most extreme measurement inside the inner fence. Thus,

$$\text{Lower inner fence} = Q_L - 1.5(\text{IQR})$$
$$\text{Upper inner fence} = Q_U + 1.5(\text{IQR})$$

3. A second pair of fences, the **outer fences**, appears at a distance of 3(IQR) from the hinges. One symbol (e.g., "*") is used to represent measurements falling between the inner and outer fences, and another (e.g., "0") is used to represent measurements that lie beyond the outer fences. Thus, outer fences are not shown unless one or more measurements lie beyond them. We have

$$\text{Lower outer fence} = Q_L - 3(\text{IQR})$$
$$\text{Upper outer fence} = Q_U + 3(\text{IQR})$$

4. The symbols used to represent the median and the extreme data points (those beyond the fences) will vary with the software you use to construct the box plot. (You may use your own symbols if you are constructing a box plot by hand.) You should consult the program's documentation to determine exactly which symbols are used.

Definition

Aids to the Interpretation of Box Plots

1. The line (median) inside the box represents the "center" of the distribution of data.
2. Examine the length of the box. The IQR is a measure of the sample's variability and is especially useful for the comparison of two samples. (See Example 2.16.)
3. Visually compare the lengths of the whiskers. If one is clearly longer, the distribution of the data is probably skewed in the direction of the longer whisker.
4. Analyze any measurements that lie beyond the fences. Less than 5% should fall beyond the inner fences, even for very skewed distributions. Measurements beyond the outer fences are probably outliers, with one of the following explanations:

   a. The measurement is incorrect. It may have been observed, recorded, or entered into the computer incorrectly.
   b. The measurement belongs to a population different from the population that the rest of the sample was drawn from. (See Example 2.17.)
   c. The measurement is correct *and* from the same population as the rest of the sample. Generally, we accept this explanation only after carefully ruling out all others.

Example 2.16

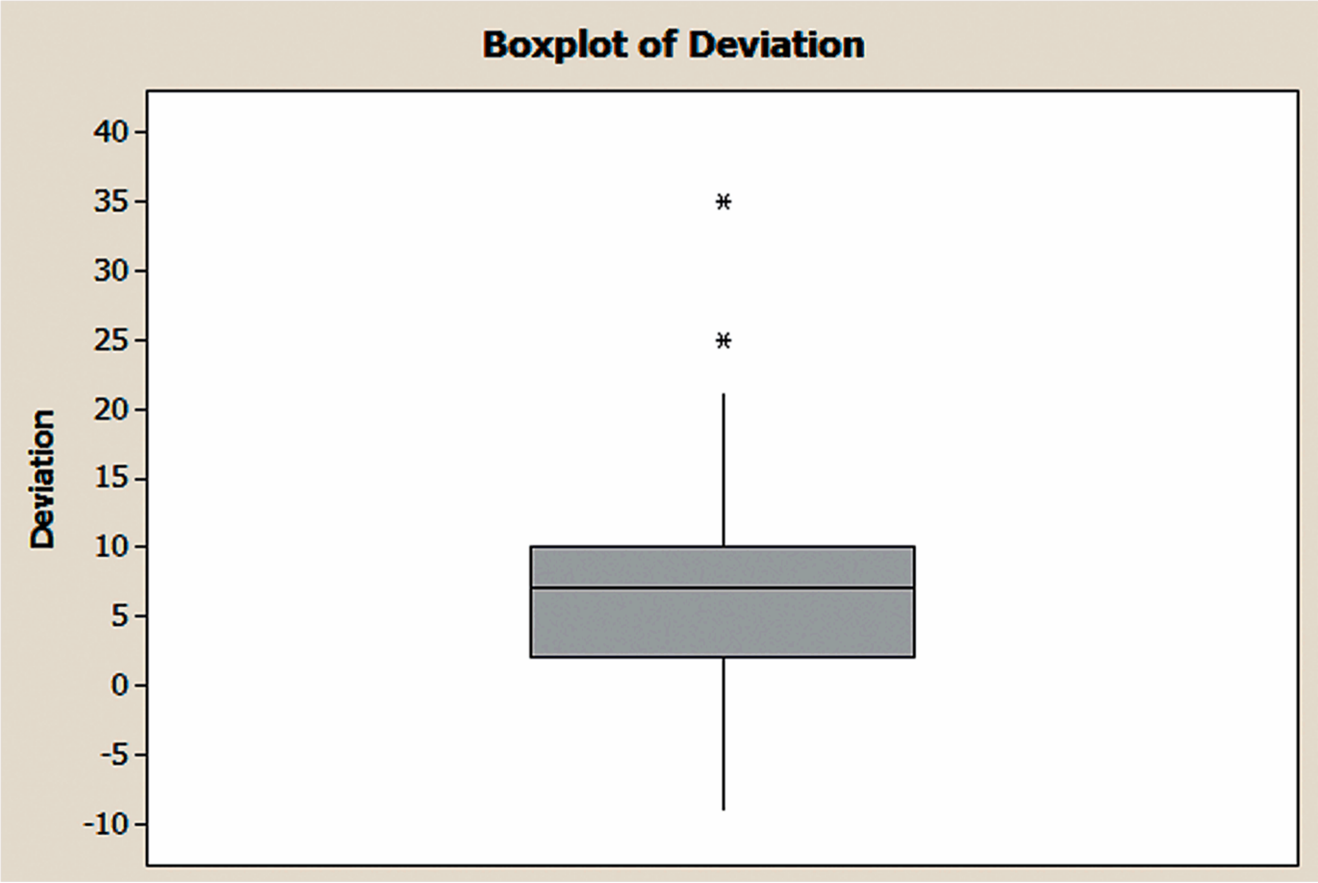
Problem Refer to the Psychological Science experiment, called the "water-level task," Examples 2.2 and 2.14. Use statistical software to produce a box plot for the 40 water-level task deviation angles shown in Table 2.4. Identify any outliers in the data set.

Table 2.4 Water-Level Task Deviations (angle degrees)

Bartenders:	−9	6	10	6	10	−3	7	8	6	14	7	8	−5	2	−1	0	2	3	0	2
Waitresses:	7	10	25	8	10	8	12	9	35	10	12	11	7	10	21	−1	4	0	16	−1

Solution - Figure 2.32 MINITAB box plot for water-level task deviations

- Note that there are two extreme observations that are **beyond the upper inner fence, but inside the upper outer fence.**
- These two values are considered **outliers**.



PEARSON

Example 2.17

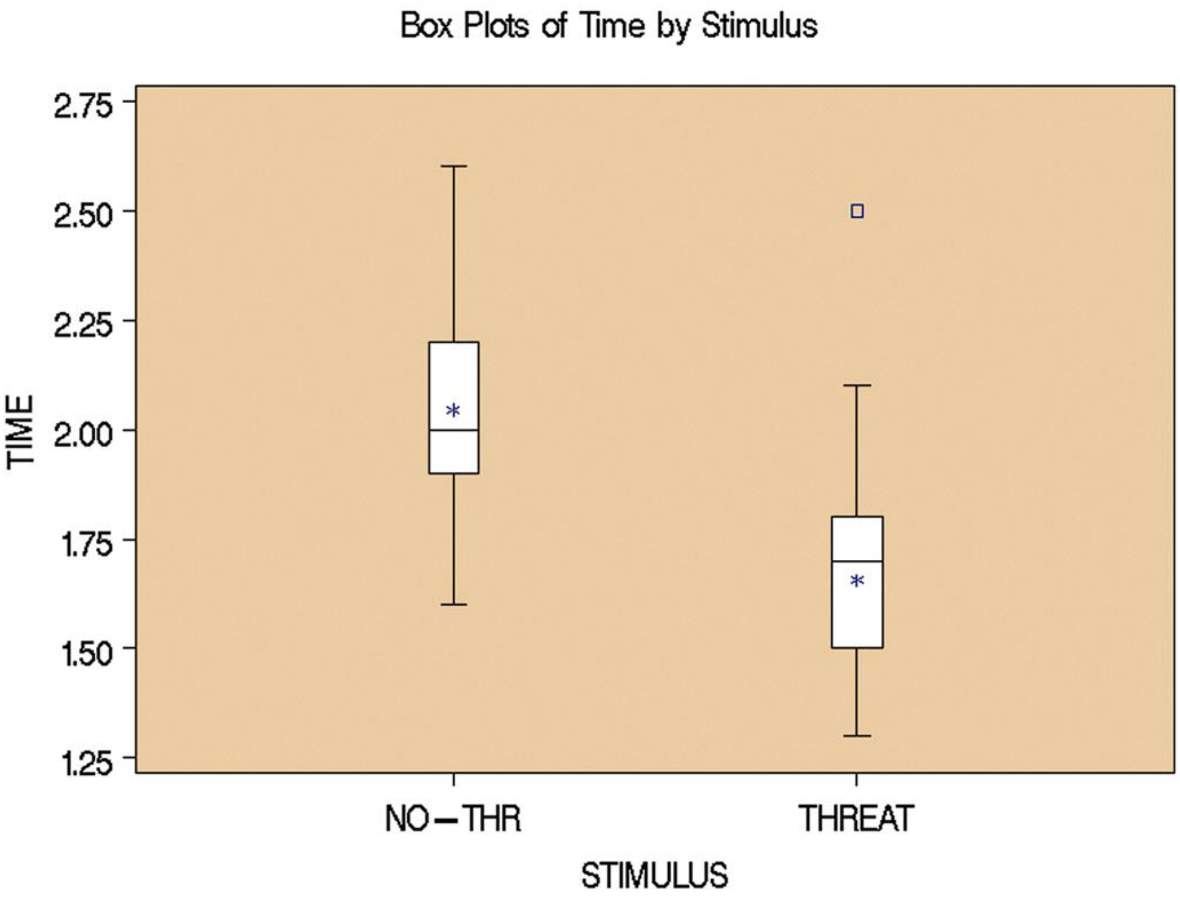
Problem A Ph.D. student in psychology conducted a stimulus reaction experiment as a part of her dissertation research. She subjected **50 subjects** to a **threatening stimulus** and **50** to a **nonthreatening stimulus**. The reaction times of all 100 students, recorded to the nearest tenth of a second, are listed in Table 2.8. Box plots of the two resulting samples of reaction times, generated with SAS, are shown in Figure 2.33. Interpret the box plots.

Table 2.8

Table 2.8 Reaction Times of Students

Nonthreatening Stimulus									
2.0	1.8	2.3	2.1	2.0	2.2	2.1	2.2	2.1	2.1
2.0	2.0	1.8	1.9	2.2	2.0	2.2	2.4	2.1	2.0
2.2	2.1	2.2	1.9	1.7	2.0	2.0	2.3	2.1	1.9
2.0	2.2	1.6	2.1	2.3	2.0	2.0	2.0	2.2	2.6
2.0	2.0	1.9	1.9	2.2	2.3	1.8	1.7	1.7	1.8
Threatening Stimulus									
1.8	1.7	1.4	2.1	1.3	1.5	1.6	1.8	1.5	1.4
1.4	2.0	1.5	1.8	1.4	1.7	1.7	1.7	1.4	1.9
1.9	1.7	1.6	2.5	1.6	1.6	1.8	1.7	1.9	1.9
1.5	1.8	1.6	1.9	1.3	1.5	1.6	1.5	1.6	1.5
1.3	1.7	1.3	1.7	1.7	1.8	1.6	1.7	1.7	1.7

Figure 2.33 SAS box plots for reaction-time data



Example 2.18

Problem Suppose a female bank employee believes that her salary is low as a result of sex discrimination. To substantiate her belief, she collects information on the salaries of her male counterparts in the banking business. She finds that their salaries have a **mean of $64,000** and a **standard deviation of $2,000**. Her salary is **$57,000**.

Does this information support her claim of sex discrimination?

Solution

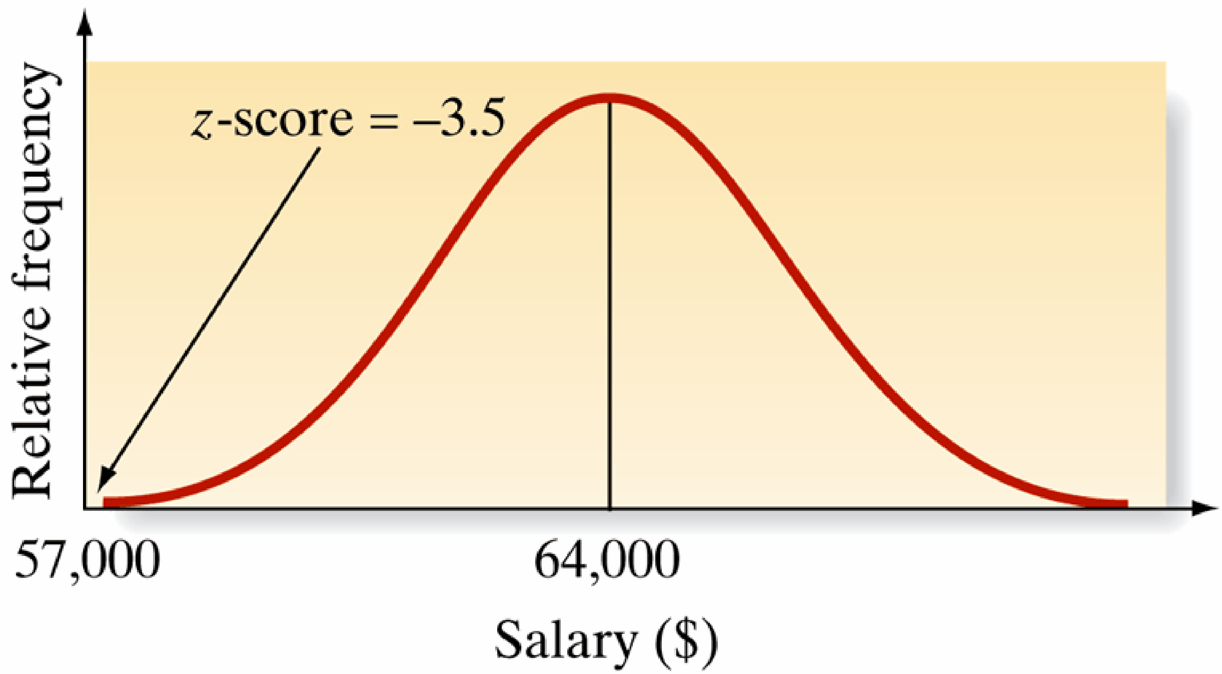
The analysis might proceed as follows: First, we calculate the z-score for the woman's salary with respect to those of her male counterparts. Thus,

$$z = \frac{\$57{,}000 - \$64{,}000}{\$2{,}000} = -3.5$$

The implication is that the woman's salary is 3.5 standard deviations below the mean of the male salary distribution. Furthermore, if a check of the male salary data shows that the frequency distribution is mound shaped, we can infer that very few salaries in this distribution should have a z-score less than -3, as shown in Figure 2.34.

Figure 2.34 Male salary distribution

Clearly, a *z*-score of -3.5 represents an outlier. Either her salary is from a distribution different from the male salary distribution, or it is a very unusual (highly improbable) measurement from a salary distribution no different from the male distribution.



PEARSON

Procedure

Rules of Thumb for Detecting Outliers[*]

Box Plots: Observations falling between the inner and outer fences are deemed *suspect outliers*. Observations falling beyond the outer fence are deemed *highly suspect outliers*.

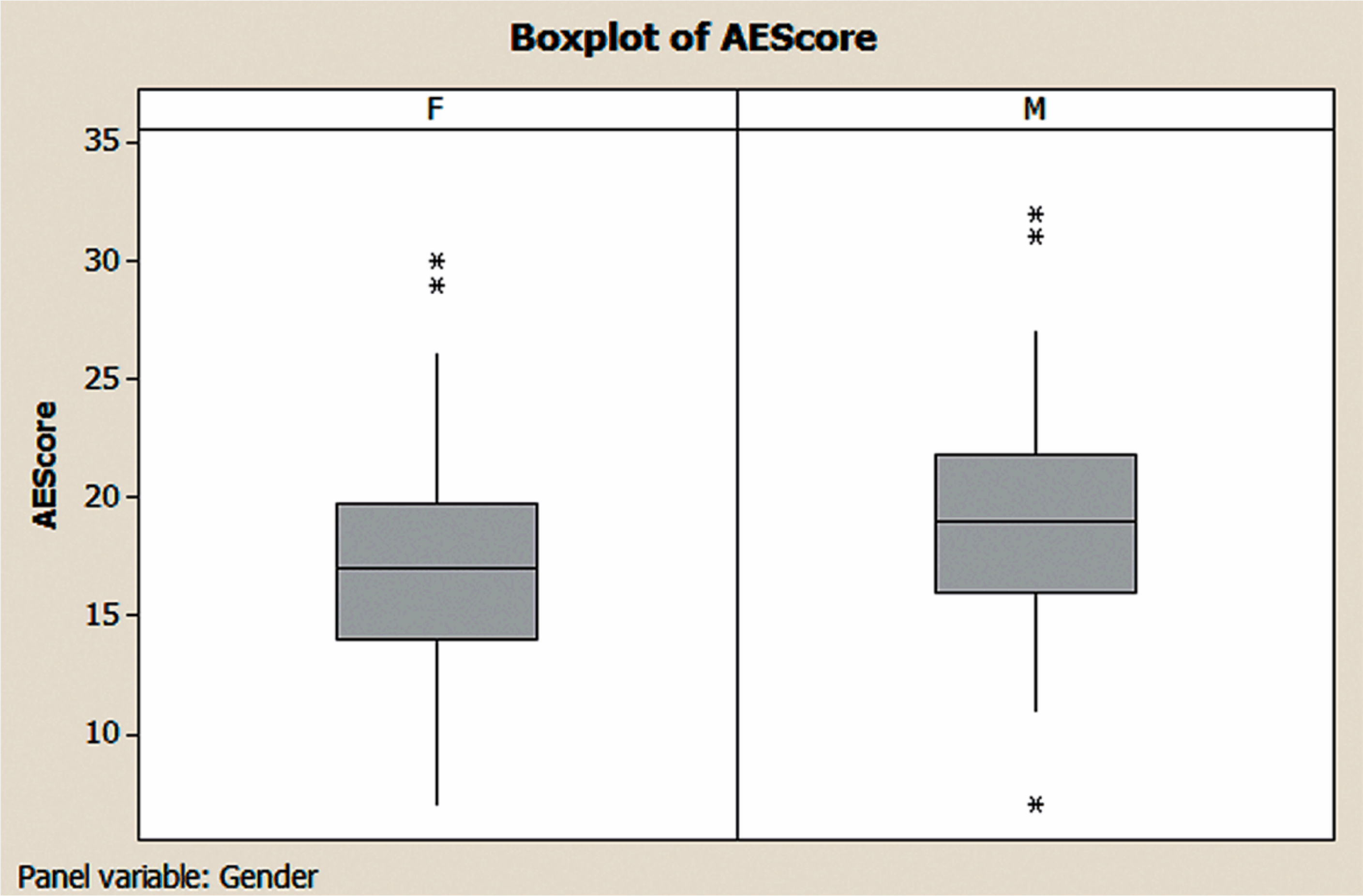
Suspect outliers	*Highly suspect outliers*
Between Q_L ... $-1.5(\text{IQR})$ and Q_L ... $-3(\text{IQR})$	Below Q_L ... $-3(\text{IQR})$
or	or
Between Q_U ... $+1.5(\text{IQR})$ and Q_U ... $+3(\text{IQR})$	Above Q_U ... $+3(\text{IQR})$

***z*-Scores:** Observations with z-scores greater than 3 in absolute value are considered *outliers*. For some highly skewed data sets, observations with z-scores greater than 2 in absolute value *may be outliers*.

Possible Outliers	*Outliers*
$\lvert z \rvert > 2$	$\lvert z \rvert > 3$

[*]The z-score and box plot methods both establish rule-of-thumb limits outside of which a measurement is deemed to be an outlier. Usually, the two methods produce similar results. However, the presence of one or more outliers in a data set can inflate the computed value of s. Consequently, it will be less likely that an errant observation would have a z-score larger than $|3|$. In contrast, the values of the quartiles used to calculate the intervals for a box plot are not affected by the presence of outliers.

Figure SIA2.6 MINITAB box plots for Appearance Evaluation score by Gender



2.8

Graphing Bivariate Relationships (Optional)

Figure 2.35 Scatterplot of cost vs. floor area

- The crime rate and the unemployment rate are “highly correlated.”
- Smoking and lung cancer are “related.”
- Words ***correlated***, ***related***, and ***associated*** imply a relationship between two variables.
- One way to describe the relationship between two quantitative variables—called a **bivariate relationship**—is to construct a **scatterplot**

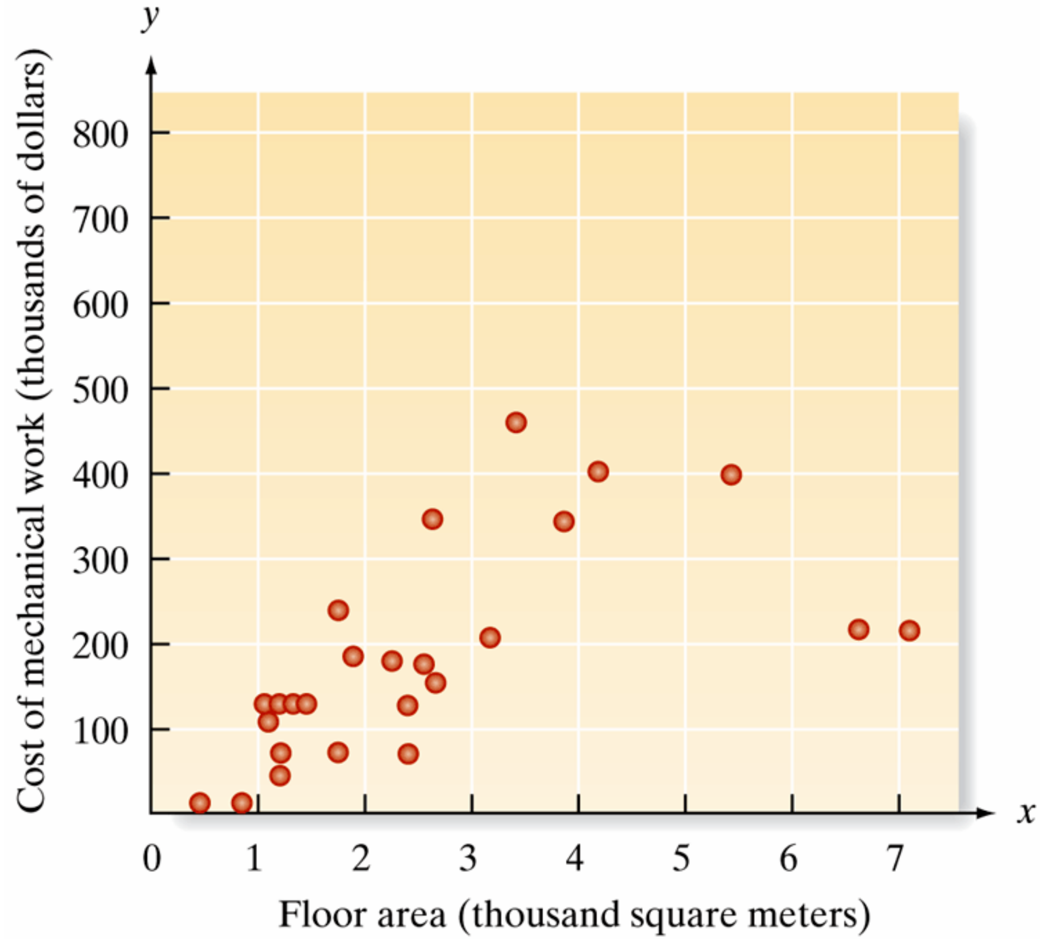
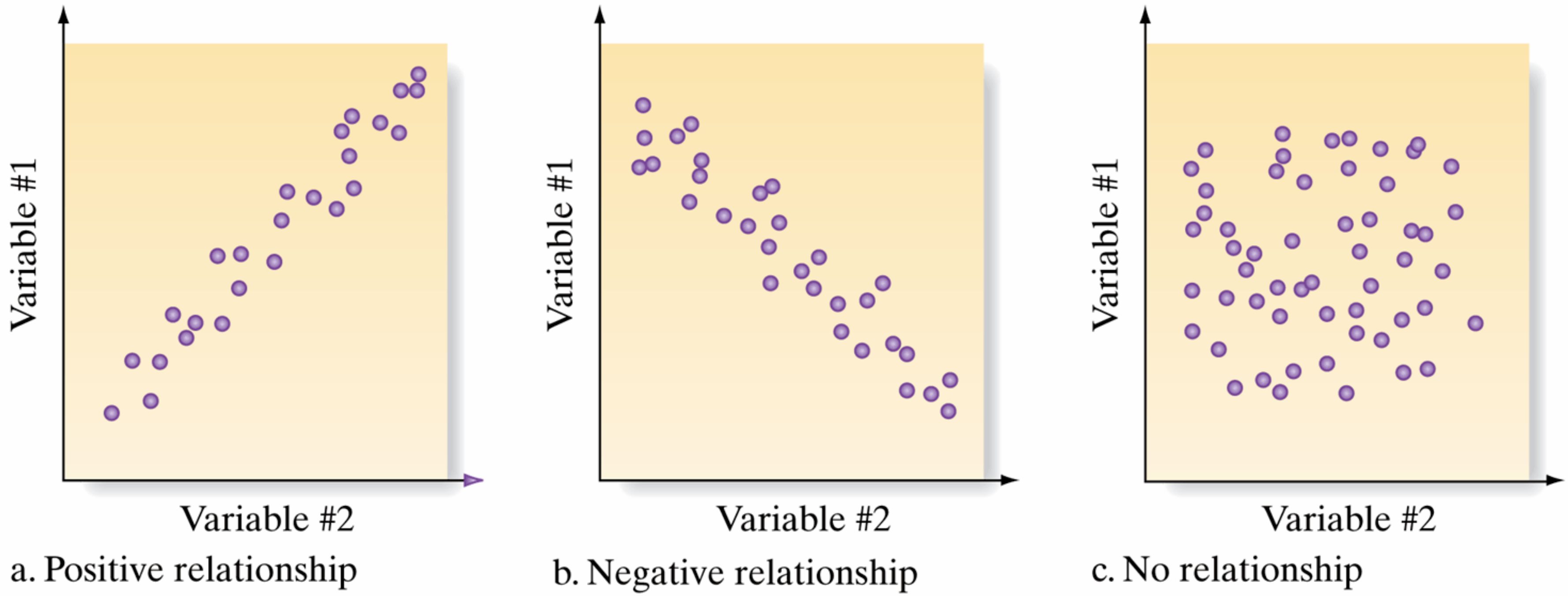


Figure 2.36 Hypothetical bivariate relationship



PEARSON

Table 2.9

A medical item used to administer to a hospital patient is called a *factor*

Use a scatterplot to describe the relationship between the two variables of interest: number of factors and length of stay.

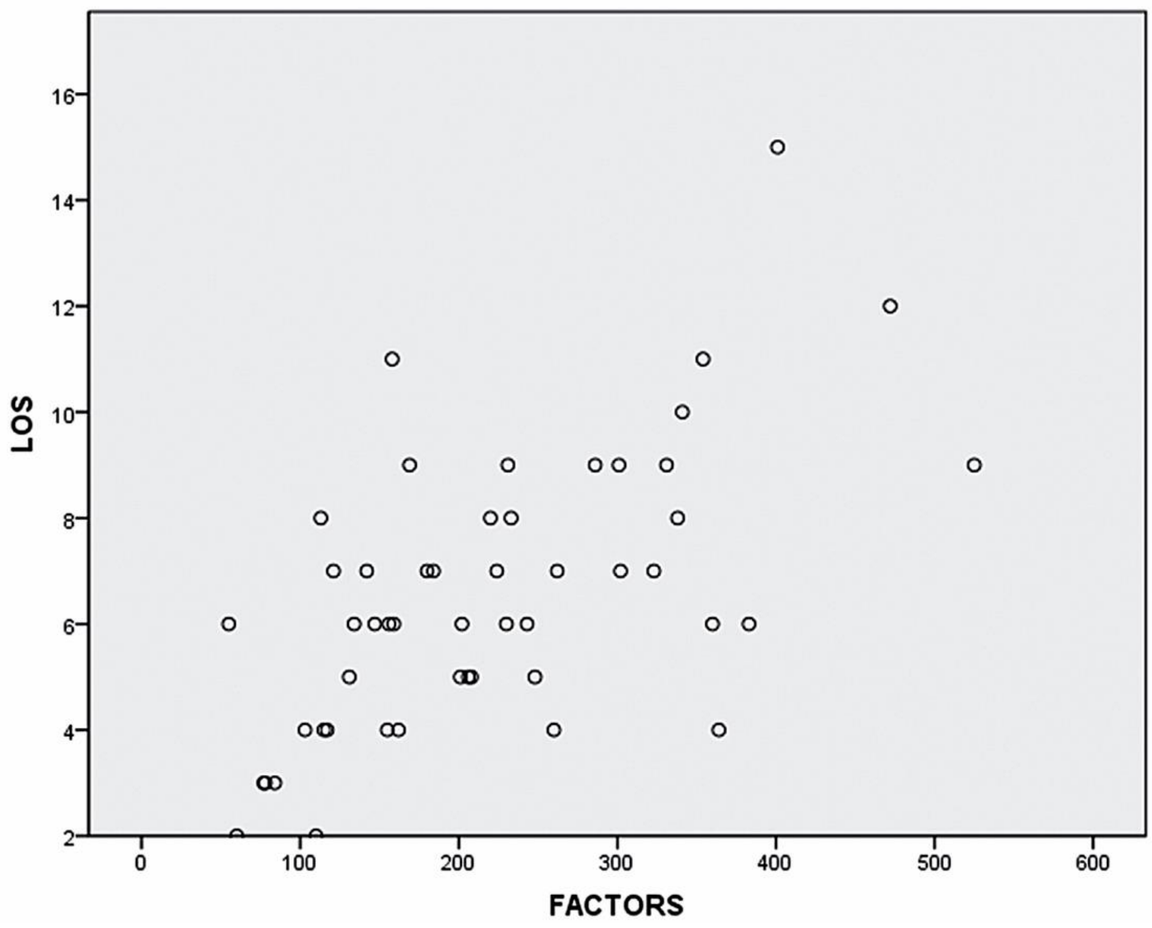
Table 2.9 Medfactors Data on Patients' Factors and Length of Stay

Number of Factors	Length of Stay (days)	Number of Factors	Length of Stay (days)
231	9	354	11
323	7	142	7
113	8	286	9
208	5	341	10
162	4	201	5
117	4	158	11
159	6	243	6
169	9	156	6
55	6	184	7
77	3	115	4
103	4	202	6
147	6	206	5
230	6	360	6
78	3	84	3
525	9	331	9
121	7	302	7
248	5	60	2
233	8	110	2
260	4	131	5
224	7	364	4
472	12	180	7
220	8	134	6
383	6	401	15
301	9	155	4
262	7	338	8

Based on Bayonet Point Hospital, Coronary Care Unit.

PEARSON

Figure 2.37 SPSS scatterplot for medical factors data in Table 2.9



PEARSON

2.9

Distorting the Truth with Descriptive Statistics

Table 2.10

Table 2.10 Collisions of Marine Vessels by Location	
Location	Number of Ships
At Sea	376
In Restricted Waters	273
In Port	478
Total	1,127

Based on *The Dock and Harbour Authority*.

Figure 2.38 MINITAB bar graph of vessel collisions by location

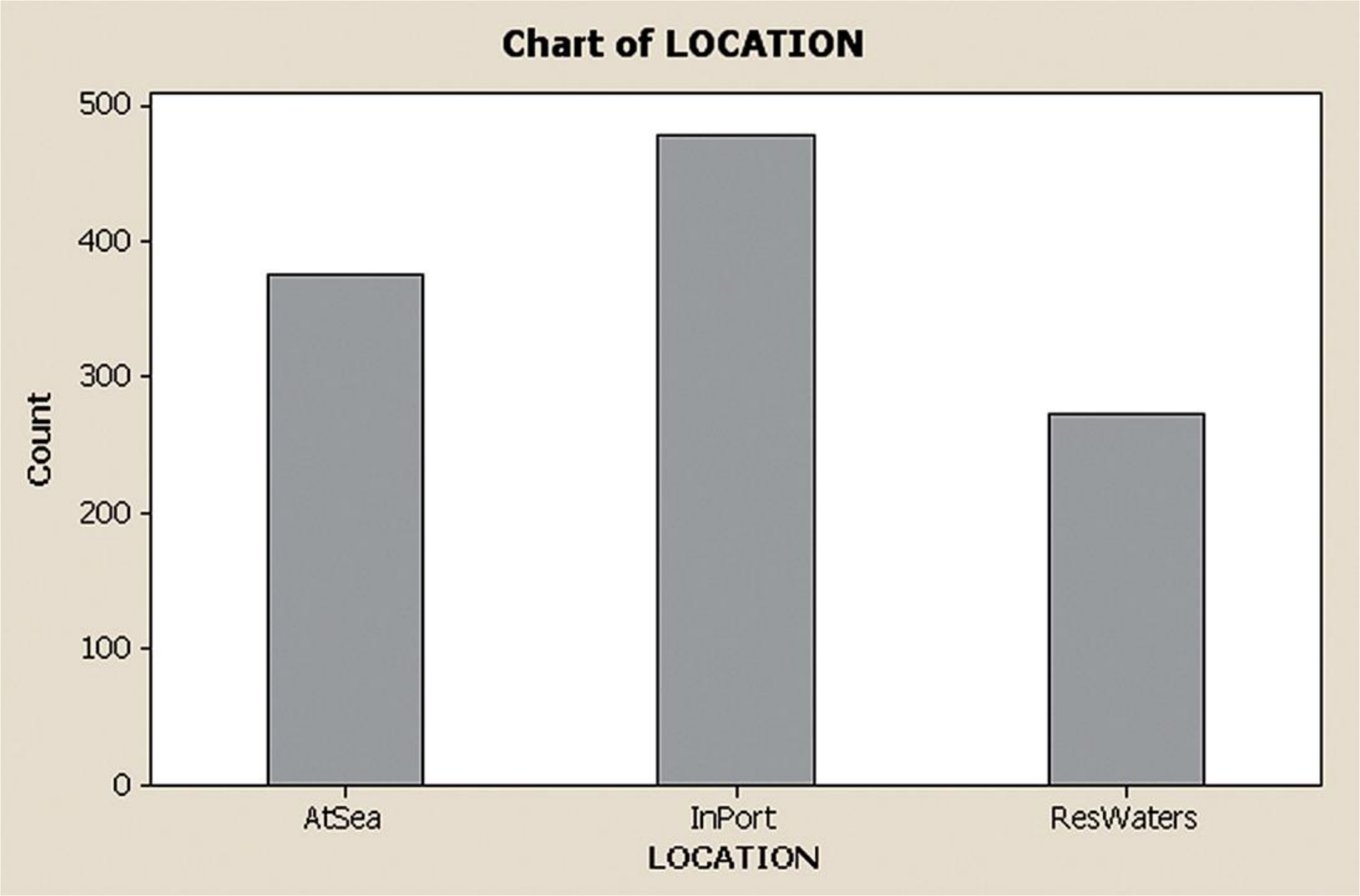


Figure 2.39 SPSS bar graph of vessel collisions by location, with adjusted vertical axis

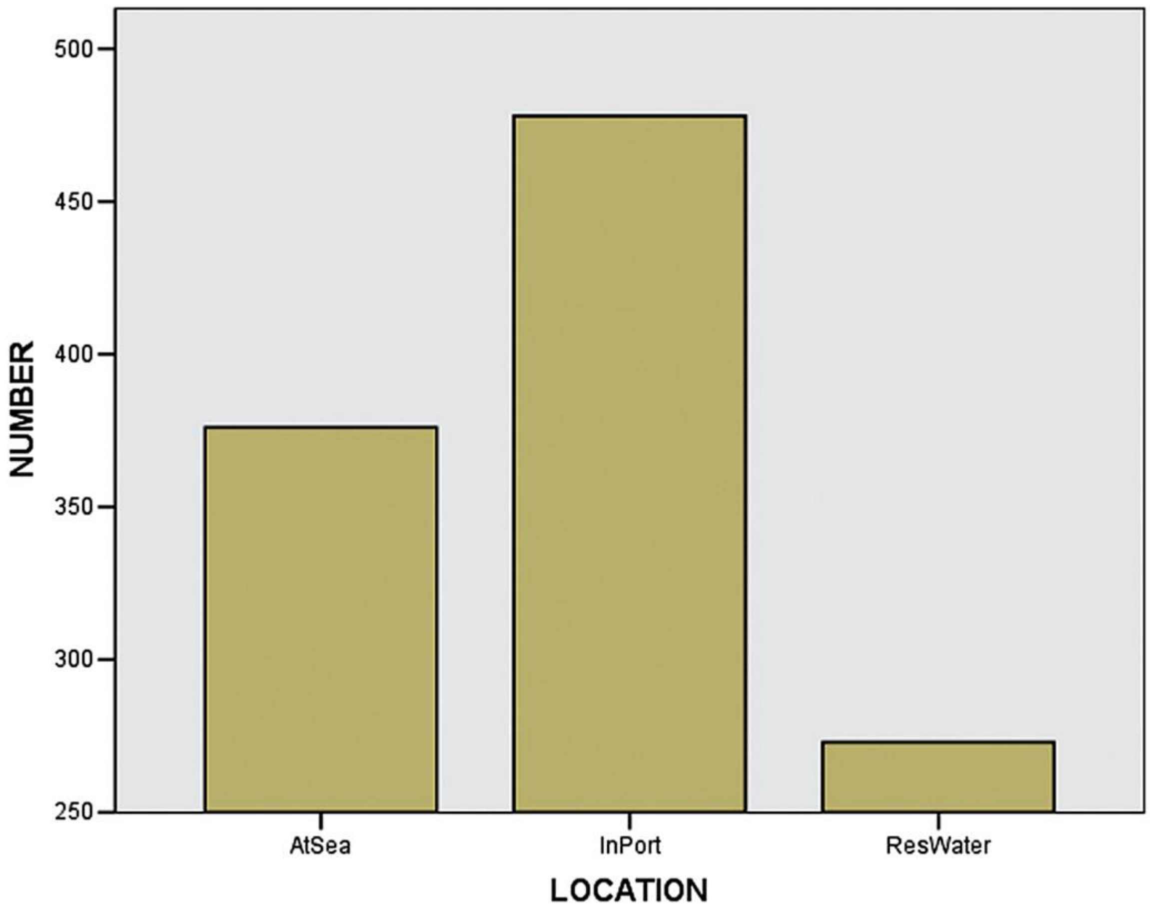


Figure 2.40 Relative frequency of fatal motor vehicle accidents on each of four major highways

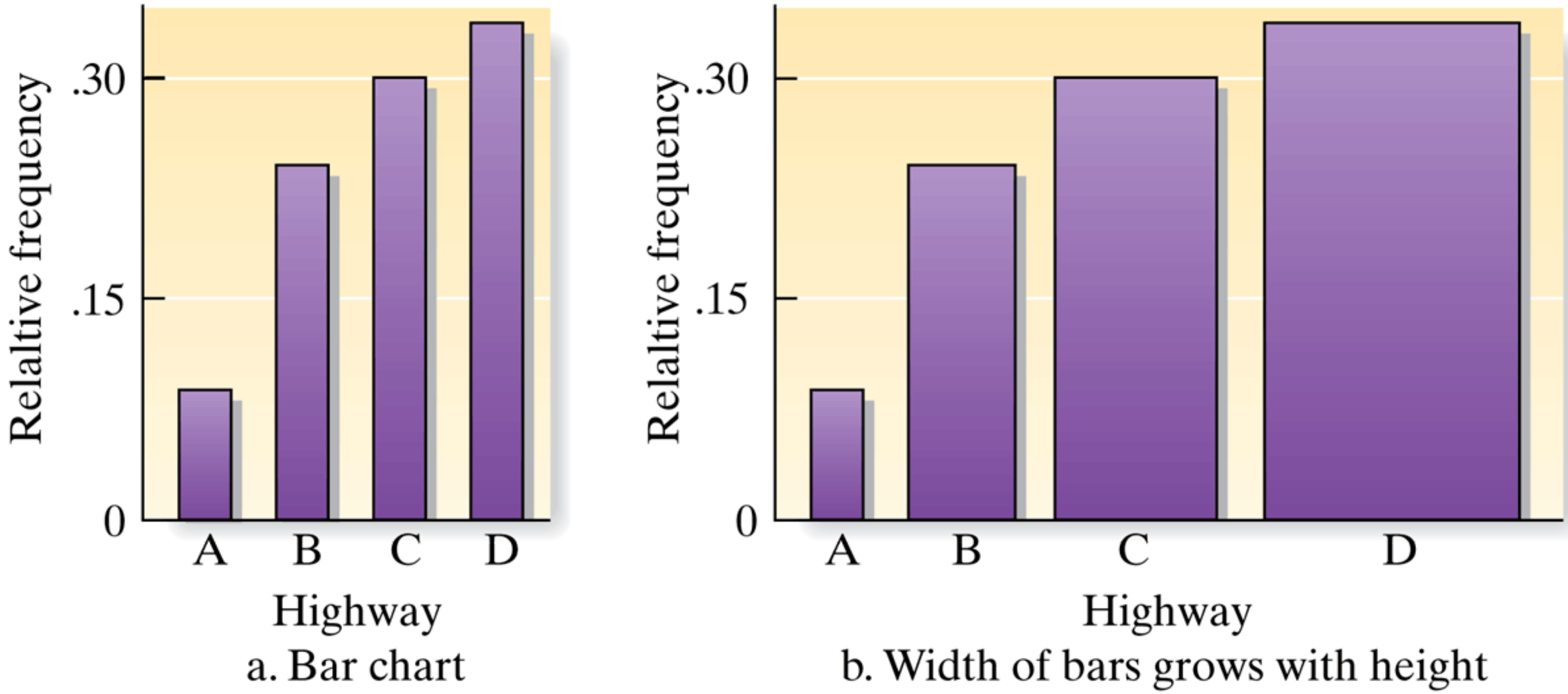
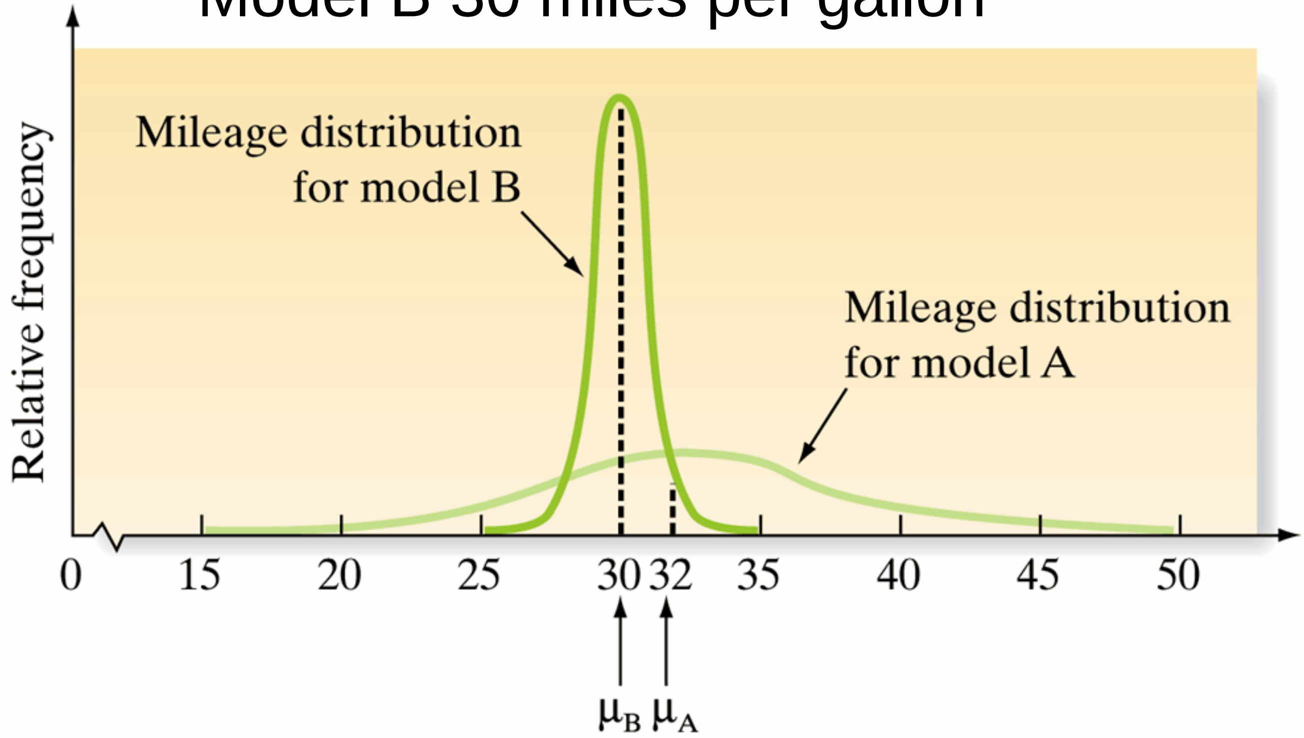


Figure 2.41 Mileage distributions for two car models

Which car would you buy?
Model A 32 miles per gallon
Model B 30 miles per gallon



PEARSON